0. Mathematical Language, Proofs & Limits

Sets, quantifiers, functions, proof techniques, limits, suprema, compactness and O-notation

BEFORE YOU START

Start here with ordinary arithmetic and school algebra. The reminders below help you recover the rest:

- Signed numbers, fractions, powers and solving simple equations: start with the warm-up.
- Reading a function such as $f(x) = x^2$: refreshed in Section 2.
- Finite sums and limits: introduced with examples in Section 4.
- Vector and derivative notation occurs in later applications. You can follow the scalar examples first; Primer A and Primer B include first-course refreshers before using those tools.

No earlier primer is needed. Read Sections 1–3 in order; return to the application boxes after the definitions feel familiar.

The modules speak the language of analysis: "for every state in C there exists an input", "$\sup_x \|J_f(x)\|$", "a compact sublevel set", "regret $O(\sqrt{T})$". This primer makes that language precise and trains the proof patterns the modules repeat: induction over time steps, chains of inequalities, geometric series, and contradiction plus compactness.

A ROUTE THROUGH THIS PRIMER

All seven sections have local graded practice: **39 exercises** in total, plus seven mixed review exercises below. Sections 1–3 offer three questions at each difficulty; Sections 4–7 offer an Easy, Medium and Hard sequence. **Easy** refreshes notation and calculations; **Medium** applies a definition or proof technique; **Hard** combines ideas and checks assumptions. Each exercise has a separate hint and a full worked answer.

Start with the Easy exercises in one section. Try each question before opening its hint, and write your reasoning before checking the answer. After checking, close the answer and try the question again from memory.

Jump to sets & logic, functions & fixed points, proofs, sequences, bounds, topology, or rates.

NOTATION — USED THROUGHOUT THIS PAGE

$\mathbb{N} = \{0, 1, 2, \dots\}$ (time starts at 0); $\|x\| = \sqrt{x^\top x}$ and $d(x, y) = \|x - y\|$ on $\mathbb{R}^n$; intervals $[a, b]$ closed, (a, b) open. t is a time step, k an iteration counter, T a horizon, $\gamma \in [0, 1)$ a discount factor. $C = \{x : h(x) \geq 0\}$ is a safe set with barrier function h (in the safe-BO modules, and in the SafeOpt examples of Sections 2, 3 and 5, h is instead a scalar safety threshold), and $x_{t+1} = F(x_t)$ a closed loop (the modules' $f(x, \pi(x))$). ∇h is the gradient of h (the vector of its partial derivatives) and J_f the Jacobian of a vector-valued f (the matrix of its partial derivatives), both from Primer B. $:=$ means "is defined as"; ■ ends a proof.

Warm-up: the algebra behind the examples

If a calculation is slowing you down, rebuild it one operation at a time. The examples below are enough to start the set and function exercises; you do not need calculus for Sections 1–3.

Background — Signs, fractions, powers and inequalities ▾

Signs and squares: $(-3)^2 = (-3)(-3) = 9$, whereas $-3^2 = -(3^2) = -9$. Absolute value discards the sign: $|-3| = 3$. Thus the distance between -2 and 1 is $|-2 - 1| = 3$.

Fractions: $\frac{1}{2} + \frac{1}{3} = \frac{3}{6} + \frac{2}{6} = \frac{5}{6}$. Dividing by a nonzero fraction multiplies by its reciprocal: $2/(1/2) = 4$. For $t \geq 0$, $1/(t + 1)$ is defined and positive; increasing its denominator decreases the value.

Powers: $2^{-3} = 1/2^3 = 1/8$, $a^{m+n} = a^m a^n$, and $a^0 = 1$ for $a \neq 0$. In particular, $0.5^0 + 0.5^1 + 0.5^2 = 1 + 1/2 + 1/4 = 7/4$. The index under a summation sign tells you where to start: $\sum_{t=0}^{2} 0.5^t$ means precisely those three terms.

Equations and inequalities: $2x + 1 = 5$ gives $2x = 4$ and then $x = 2$. Subtracting the same quantity from both sides preserves an inequality. Multiplying or dividing by a negative number reverses it: $-2x \leq 4$ gives $x \geq -2$. Hence $x^2 \leq 4$ means $-2 \leq x \leq 2$, with both signs included.

A quick self-check: before closing this box, reconstruct $(-2)^2 = 4$, $\frac{1}{4} + \frac{1}{2} = \frac{3}{4}$, $2^{-2} = \frac{1}{4}$ and $-3x > 6 \iff x < -2$. If one line is unfamiliar, use its rule on a new number and check by substitution.

1. Sets, Quantifiers & Logic

Every safety statement in the modules has the same grammar: a set of states or parameters, a test its points must pass, and quantifiers saying whether one point, every point or every time step must pass it.

DEFINITION — SET, SUBSET, SET-BUILDER NOTATION

A **set** is a collection of distinct **elements**; $x \in A$ means that x is an element of A, and $x \notin A$ that it is not. $A \subseteq B$ (A is a **subset** of B) means that every element of A lies in B, and $A = B$ iff $A \subseteq B$ and $B \subseteq A$; $\emptyset$ is the empty set. **Set-builder notation** $\{x \in X : \ P(x)\}$ denotes the elements x of an ambient set X for which the statement $P(x)$ is true. Read it as a filter: run through X and keep x if it passes the test P. (Some papers and modules write $\subset$ for $\subseteq$; this page writes $\subsetneq$ for a proper subset.)

Example. For $h(x) = 1 - x_1^2 - x_2^2$ the safe set $C = \{x \in \mathbb{R}^2 : h(x) \geq 0\}$ is the closed unit disc. $(0.6, 0.8) \in C$, since $h = 1 - 0.36 - 0.64 = 0$ (it lies on the edge), while $(1, 1) \notin C$, since $h = -1$. Also $C \subseteq [-1, 1]^2$, because $x \in C$ forces $x_i^2 \leq 1$; the reverse inclusion fails, as $(1, 1)$ shows.

DEFINITION — OPERATIONS ON SETS

For $A, B \subseteq X$: $A \cup B$ (in A or B or both), $A \cap B$ (in both), $A \setminus B = \{x \in A : x \notin B\}$, the **complement** $A^c = X \setminus A$ (it depends on the ambient set X), the **product** $A \times B = \{(a, b) : a \in A, \ b \in B\}$ of ordered pairs, the **cardinality** $|A|$ (number of elements of a finite A), and the set 2^X of all subsets of X. For sets A_i indexed by $i \in I$:

$$\bigcup_{i \in I} A_i = \{x : \ x \in A_i \text{ for at least one } i\}, \qquad \bigcap_{i \in I} A_i = \{x : \ x \in A_i \text{ for every } i\}.$$

De Morgan's laws: $\left(\bigcup_i A_i\right)^c = \bigcap_i A_i^c$ and $\left(\bigcap_i A_i\right)^c = \bigcup_i A_i^c$. *Proof.* $x \notin \bigcup_i A_i$ iff no i has $x \in A_i$ iff $x \in A_i^c$ for every i; applying this to the sets A_i^c and taking complements gives the second law. ■

Example. With $D = \{1, \ldots, 6\}$, $S = \{2, 3\}$, $T = \{3, 4, 5\}$: $S \cup T = \{2, 3, 4, 5\}$, $S \cap T = \{3\}$, $|D \setminus S| = |\{1, 4, 5, 6\}| = 4$, and $S \times \{1, 2\}$ has 4 pairs. Several constraints form an intersection, $\bigcap_i \{a : g_i(a) \geq 0\}$; "a failure at some $t \leq T$" is a union, $\bigcup_{t=0}^{T} \{x_t \notin C\}$.

WHERE THIS IS USED

Safe sets in Module 1; seeds $S_0 \subseteq D$ and indexed unions in Module 4, pairs $A \times I$ in Module 4 and Module 6; $\{a : g_i(a) \geq 0 \ \forall i\}$ and $|S|$ in Module 6; $X \times U$ and $X \setminus X_F$ in Module 7; intersected intervals $C_t(x) = C_{t-1}(x) \cap Q_t(x)$ in Module 5.

Statements and implications

DEFINITION — CONNECTIVES, CONVERSE, CONTRAPOSITIVE

For statements P, Q (each true or false): $\neg P$ (not P), $P \wedge Q$ (and), $P \vee Q$ (or, including both), $P \Rightarrow Q$ (if P then Q; false only when P is true and Q false), and $P \Leftrightarrow Q$ (both directions). The **converse** of $P \Rightarrow Q$ is $Q \Rightarrow P$; its **contrapositive** is $\neg Q \Rightarrow \neg P$. An implication is equivalent to its contrapositive and to $\neg P \vee Q$, but not to its converse. "P is **sufficient** for Q" means $P \Rightarrow Q$; "P is **necessary** for Q" means $Q \Rightarrow P$.

Intuition and example. An implication is a promise, broken only when P holds and Q fails; if P never holds it is *vacuously* true, so a condition checked on an empty set certifies nothing. "$x > 2 \Rightarrow x^2 > 4$" is true; its converse is false ($x = -3$); its contrapositive "$x^2 \leq 4 \Rightarrow x \leq 2$" is true. Module 14 requires $\big(D(\xi) \leq 0 \ \wedge \ f_{\mathrm{cl}}(\xi) \in \mathcal{B}\big) \vee V(\xi) \geq \rho$ for all $\xi \in \mathcal{B}$, with $D(\xi) \leq 0$ a decrease condition on V (the module writes F). Since $P \vee Q \equiv (\neg Q \Rightarrow P)$: whenever $V(\xi) < \rho$, V decreases and the successor $f_{\mathrm{cl}}(\xi)$ stays in $\mathcal{B}$.

COMMON MISTAKE — READING A CERTIFICATE BACKWARDS

Certificates are sufficient conditions ("the LMI is feasible $\Rightarrow$ the loop is stable"), and a failed certificate proves nothing: "infeasible $\Rightarrow$ unstable" is the *inverse* $\neg P \Rightarrow \neg Q$, which is equivalent to the converse, not to the statement. The S-procedure of Module 2 is special because, for one constraint (and a strictly feasible point), its sufficient condition is also necessary.

Quantifiers, negation, and why their order matters

DEFINITION — QUANTIFIERS AND THEIR NEGATION

"$\forall x \in X:\ P(x)$" (for all) is true iff every element of X satisfies P; it is vacuously true if $X = \emptyset$. "$\exists x \in X:\ P(x)$" (there exists) is true iff at least one element does. Negation swaps them:

$$\neg\big(\forall x \in X : P(x)\big) \iff \exists x \in X : \neg P(x), \qquad \neg\big(\exists x \in X : P(x)\big) \iff \forall x \in X$$

To negate a chain of quantifiers, swap every $\forall \leftrightarrow \exists$ and negate the innermost statement. In set language, "for every i" is an intersection and "for some i" a union.

Intuition. In "$\forall x\ \exists y : P(x,y)$" you may answer each x with its own y; "$\exists y\ \forall x : P(x,y)$" forces one y before seeing x. Committing first is harder: $\exists y\,\forall x\ P$ implies $\forall x\,\exists y\ P$, but not conversely. Over $\mathbb{R}$, "$\forall x\ \exists y : y > x$" is true (answer $y = x + 1$), while "$\exists y\ \forall x : y > x$" claims a largest real number and is false.

Worked example — one witness for all constraints, or one per constraint (Module 6) ▾

A parameter a is certified from evaluated parameters $a' \in S$ through a Lipschitz bound L on constraint functions g_i (larger is safer). Module 6 compares two rules that differ only in quantifier order (the module also subtracts an accuracy ϵ and adds S itself):

$$\exists a' \in S\ \forall i:\ g_i(a') - L\|a - a'\| \geq 0 \qquad \text{versus} \qquad \forall i\ \exists a'_i \in S:\ g_i(a'_i) - L\|a - a'_i\| \geq$$

Take $S = \{0, 2\}$, $a = 1$, $L = 1$ (both witnesses at distance 1), $g_1(0) = 2$, $g_1(2) = 0.5$, $g_2(0) = 0.5$, $g_2(2) = 2$. Witness 0 certifies constraint 1 ($2 - 1 \geq 0$) but not constraint 2 ($0.5 - 1 < 0$), and witness 2 the opposite. No single witness works, so a fails the first rule, but $a'_1 = 0$, $a'_2 = 2$ satisfy the second. Since "$\exists\forall$" implies "$\forall\exists$", the per-constraint rule always certifies at least as much.

Worked example — negating the viability kernel (Modules 1 and 7) ▾

With $\mathcal{U}$ the admissible control signals and $\varphi_x^u(t)$ the state at time t from x under u, the viability kernel of Module 7 is $X_V = \{x:\ \exists u \in \mathcal{U}\ \forall t \geq 0:\ \varphi_x^u(t) \notin X_F\}$: some control avoids the failure set X_F forever. Swap the quantifiers and negate the inside: $X \setminus X_V = \{x:\ \forall u \in \mathcal{U}\ \exists t \geq$

$0 : \ \varphi_x^u(t) \in X_F\}$, every control fails at some time, which is the unviability kernel X_U of Module 1. The failure time t may depend on u.

Order inside probabilities. "$\forall t : \ \mathbb{P}(x_t \in C) \geq 1 - \delta$" is much weaker than "$\mathbb{P}(\forall t : \ x_t \in C) \geq 1 - \delta$": with independent 5% failure chances per step, all 20 steps are safe only with probability $0.95^{20} \approx 0.358$. The union bound $\mathbb{P}(\exists t \leq T : x_t \notin C) \leq \sum_{t=0}^{T} \delta_t$ connects the two (Primer C).

WHERE THIS IS USED

Both certification rules in Module 6; "$\exists$ control $\forall$ time" in Module 7 and its complement in Module 1; the alternating quantifiers $\exists \beta \ \forall a(\cdot) \ \exists s$ of a reach-avoid game in Module 10; $\mathbb{P}(\exists t : x_t \notin C)$ in Module 1; "for every policy some parameterised policy is within ϵ" in Module 2.

COMMON MISTAKE — NEGATING ONLY THE INSIDE

The negation of "$\forall x \in C \ \exists u \in U : f(x, u) \in C$" ($C$ is control invariant) is "$\exists x \in C \ \forall u \in U : f(x, u) \notin C$" (some state of C is lost whatever you do), not "$\forall x \ \exists u : f(x, u) \notin C$".

Reading aids. "Iff" (if and only if): prove both directions. "Without loss of generality" (WLOG): a reduction that loses no cases, by symmetry or by a change of variables. For example, an equilibrium x^* of $x_{t+1} = F(x_t)$ is moved to the origin by $z = x - x^*$ and $\tilde{F}(z) = F(z + x^*) - x^*$, which gives $\tilde{F}(0) = 0$; hence "WLOG $x^* = 0$", provided constraint sets and functions such as V are shifted too. "s.t." means "such that", or "subject to" in optimization.

Practice: Sets, Quantifiers & Logic

Start by listing elements and reading symbols aloud. The Medium exercises ask you to translate statements; the Hard exercises combine quantifiers with constraints. If a definition feels unfamiliar, revisit this section before trying again.

▼ Exercise 0.S1 — Easy: Read a set-builder expression

Let $X = \{-2, -1, 0, 1, 2, 3\}$, $A = \{x \in X : x^2 \leq 4\}$, and $B = \{x \in X : x > 0\}$. List A and B. Is $2 \in A$? Is $B \subseteq A$? Give a reason for each answer.

Review: sets and logic

▼ Show hint

Test each element of X separately. A subset claim fails if you find one element of B outside A.

▼ Show answer

> $A = \{-2, -1, 0, 1, 2\}$: their squares are $4, 1, 0, 1, 4$, while $3^2 = 9 > 4$. The positive elements give $B = \{1, 2, 3\}$.
>
> Yes, $2 \in A$, because $2^2 = 4 \leq 4$. But $B \not\subseteq A$: $3 \in B$ and $3 \notin A$. Notice the difference between membership ($2 \in A$) and inclusion (one set inside another).

▼ Exercise 0.S2 — Easy: Union, intersection, difference and complement

Use the universe $X = \{0, 1, 2, 3, 4, 5\}$ and sets $A = \{0, 2, 4\}$, $B = \{2, 3, 4\}$. Compute $A \cup B$, $A \cap B$, $A \setminus B$, and A^c.

Review: sets and logic

▼ Show hint

Union means either set; intersection means both. For A^c, keep the elements of the stated universe that are outside A.

▼ Show answer

> $A \cup B = \{0, 2, 3, 4\}$: collect every element that appears in either list, counting repeats once. $A \cap B = \{2, 4\}$: these appear in both.
>
> $A \setminus B = \{0\}$: start with A and remove $2, 4$, which also belong to B. Finally $A^c = X \setminus A = \{1, 3, 5\}$. Complements depend on the universe: this complement contains only elements of X, not every real number outside A.

▼ Exercise 0.S3 — Easy: True or false, with a witness

Decide whether each statement is true. Explain a true universal statement, give a witness for a true existential statement, and explain why a false statement fails.

1. $\forall x \in \mathbb{R}:\ x^2 \geq 0$.
2. $\exists x \in \mathbb{R}:\ x^2 = -1$.
3. For every real x there is a real y with $y = x + 2$.
4. There is one real y such that $y = x + 2$ for every real x.

Review: sets and logic

▼ Show hint

In the third statement you may choose y after seeing x. In the fourth, fix y first and try $x = y$.

▼ Show answer

1. True: a real number times itself is nonnegative, including $0^2 = 0$.
2. False: the first fact rules out a negative square.
3. True: for any given x, choose $y = x + 2$. This is a real number and satisfies the required equation.
4. False: whatever y was chosen, the allowed choice $x = y$ would require $y = y + 2$, hence $0 = 2$. A witness that can depend on x need not work for every x at once.

▼ Exercise 0.S4 — Medium: Negate every quantifier

Negate both statements, keeping the domains unchanged. Then decide whether the original statements are true.

1. $\forall x \in \mathbb{R}:\ x^2 \geq 1$.
2. There exists $x \in [0,1]$ such that $x > y$ for every $y \in [0,1]$.

Review: sets and logic

▼ Show hint

Swap every "for every" with "there exists", then negate the final comparison. The negation of a strict inequality includes equality.

▼ Show answer

> For (1), the negation is "there exists $x \in \mathbb{R}$ with $x^2 < 1$". Take $x = 0$: $0^2 = 0 < 1$, so the original is false.
>
> For (2), the negation is "for every $x \in [0,1]$ there exists $y \in [0,1]$ with $x \leq y$". Choose $y = x$ for each x. This proves the negation, so the original is false: no point is strictly greater than itself. Both the quantifier order and the inequality changed.

▼ Exercise 0.S5 — Medium: Converse, contrapositive and necessary conditions

For real x, consider P: "$x > 3$" and Q: "$x^2 > 9$". Write and assess $P \Rightarrow Q$, its converse, and its contrapositive. In the original implication, is P necessary or sufficient for Q? What is Q for P?

Review: sets and logic

▼ Show hint

Try a negative number for the converse. "P is sufficient for Q" means P guarantees Q.

▼ Show answer

> The original is $x > 3 \Rightarrow x^2 > 9$, which is true: both numbers are positive and squaring preserves their order.
>
> The converse is $x^2 > 9 \Rightarrow x > 3$. It is false: $x = -4$ has square $16 > 9$ but is not greater than 3.
>
> The contrapositive is $x^2 \leq 9 \Rightarrow x \leq 3$. It is true, because the premise gives $-3 \leq x \leq 3$. Thus P is sufficient for Q, and Q is necessary for P. The false converse shows that P is not necessary for Q.

▼ Exercise 0.S6 — Medium: An infinite union and intersection

For each integer $k \geq 1$, let $A_k = [-1/k, 1/k]$. Find $\bigcup_{k \geq 1} A_k$ and $\bigcap_{k \geq 1} A_k$. Justify why no nonzero real number lies in the intersection.

Review: sets and logic

▼ Show hint

A_1 contains every later interval. For a nonzero x, choose an integer $k > 1/|x|$.

▼ Show answer

The union is $[-1, 1]$: $A_1 = [-1, 1]$ already supplies every point in it, and every A_k is contained in A_1.

The intersection is $\{0\}$. First, $0 \in A_k$ for every k. Now let $x \neq 0$. Choose an integer $k > 1/|x|$; then $1/k < |x|$, so $x \notin A_k$. Membership in the intersection requires membership in every interval, so that one failure excludes x.

▼ Exercise 0.S7 — Hard: Prove De Morgan's law for an interval

In the universe $\mathbb{R}$, set $A = \{x : x \geq -1\}$ and $B = \{x : x \leq 2\}$. Find $(A \cap B)^c$. Prove that it equals $A^c \cup B^c$ by following an arbitrary real x through the membership conditions.

Review: sets and logic

▼ Show hint

Outside "both conditions hold" means "at least one condition fails". Pay attention to whether the endpoints remain.

▼ Show answer

$A \cap B = [-1, 2]$, so its complement is $(-\infty, -1) \cup (2, \infty)$. The endpoints -1 and 2 belong to the intersection and are excluded from its complement.

$$\begin{aligned} x \in (A \cap B)^c &\iff \neg(x \geq -1 \ \wedge \ x \leq 2) \\ &\iff x < -1 \ \vee \ x > 2 \\ &\iff x \in A^c \cup B^c. \end{aligned}$$

This holds for every real x, so the two sets contain exactly the same elements and are equal.

▼ Exercise 0.S8 — Hard: One input per state or one input for all states

Let $X = \{a, b, c\}$ be states and $U = \{L, R\}$ be inputs. A pair is safe exactly when it belongs to $S = \{(a, L), (b, R), (c, L), (c, R)\}$. Decide whether each statement is true:

1. For every $x \in X$ there exists $u \in U$ with $(x, u) \in S$.
2. There exists $u \in U$ such that $(x, u) \in S$ for every $x \in X$.

Negate statement (1) and assess its negation.

Review: sets and logic

▼ Show hint

List the permitted inputs at each state. A single input must appear in all three lists.

▼ Show answer

State a permits only L, state b only R, and state c both. Statement (1) is true: use L at a, R at b, and, for example, L at c.

Statement (2) is false. Input L fails at b, while R fails at a; these are all the possible inputs.

The negation of (1) is "there exists $x \in X$ such that for every $u \in U$, $(x, u) \notin S$". It is false: every state has at least one permitted input. Failure of a single shared input does not mean some state has no input.

▼ **Exercise 0.S9 — Hard: Find the parameter range for a quantified claim**

Let $a \geq 0$, $C = [-1, 1]$, and $U = [-a, a]$. An input u cancels a state x when $x + u = 0$. For which a is it true that every $x \in C$ has a cancelling $u \in U$? For which a can one fixed $u \in U$ cancel every $x \in C$? Prove both answers.

Review: sets and logic

▼ **Show hint**

Solve for the only possible input, $u = -x$. Use $x = 1$ to test the size of U, then compare $x = 1$ and $x = -1$ for a shared input.

▼ **Show answer**

> The first claim holds exactly when $a \geq 1$. Necessity: at $x = 1$ cancellation requires $u = -1$, which lies in $[-a, a]$ only if $a \geq 1$. Sufficiency: if $a \geq 1$, for any $x \in [-1, 1]$ choose $u = -x$. Then $|u| = |x| \leq 1 \leq a$, so $u \in U$, and $x + u = 0$.
>
> The shared-input claim is false for every $a \geq 0$, however large. Cancelling $x = 1$ requires $u = -1$, while cancelling $x = -1$ requires $u = 1$. One real input cannot equal both.

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2. Functions, Maps & Fixed Points

Controllers, networks, Bellman operators and SafeOpt's set expansion are functions, and many safety objects are their *fixed points*.

> **DEFINITION — FUNCTION, IMAGE, PREIMAGE, COMPOSITION, RESTRICTION**
>
> A **function** $f : X \to Y$ assigns to each x in its **domain** X exactly one value $f(x)$ in its **codomain** Y. For $A \subseteq X$ and $B \subseteq Y$, the **image** is $f(A) = \{f(x) : x \in A\}$ and the **preimage** is $f^{-1}(B) = \{x \in X : f(x) \in B\}$. The composition of f with $g : Y \to Z$ is $(g \circ f)(x) = g(f(x))$; the restriction $f|_A$ is f with domain A. A function of several variables lives on a product, e.g. dynamics $f : X \times U \to \mathbb{R}^n$.

Intuition and example. The image collects outputs, the preimage the inputs that land in B. Safe sets and sublevel sets are preimages: $C = \{x : h(x) \geq 0\} = h^{-1}([0, \infty))$, the zero level set is $h^{-1}(\{0\})$, and $\mathcal{V}(c) = \{x : V(x) \leq c\} = V^{-1}((-\infty, c])$. For $f(x) = x^2$: $f([-1, 2]) = [0, 4]$, $f^{-1}([1, 4]) = [-2, -1] \cup [1, 2]$, $f^{-1}([-3, -1]) = \emptyset$, and $f^{-1}(\{1\}) = \{-1, 1\}$, although f has no inverse.

FACT — PREIMAGES RESPECT SET OPERATIONS; IMAGES ONLY UNIONS

$f^{-1}(B_1 \cap B_2) = f^{-1}(B_1) \cap f^{-1}(B_2)$, likewise for $\cup$, and $f^{-1}(Y \setminus B) = X \setminus f^{-1}(B)$. *Proof of the first:* x belongs to either side iff $f(x) \in B_1$ and $f(x) \in B_2$. ■ So several constraints give $C = \bigcap_i h_i^{-1}([0, \infty))$. Images respect unions, $f(A_1 \cup A_2) = f(A_1) \cup f(A_2)$, but for intersections only $f(A_1 \cap A_2) \subseteq f(A_1) \cap f(A_2)$ holds: for $f(x) = x^2$, $A_1 = \{-1\}$ and $A_2 = \{1\}$ the left side is empty and the right side is $\{1\}$. Images introduce "there exists": the projection $\pi_X(a, x) = x$ maps a set S of pairs to $\pi_X(S) = \{x : \exists a,\ (a, x) \in S\}$.

WHERE THIS IS USED

Safe sets and equilibria $F(x^*) = x^*$ in Module 1; sublevel sets in Module 11; $h|_C$ and a domain $D \supseteq C$ in Module 10; the projection $P_X(S)$ in Module 6; slices $Q_V[x] = \{u : (x, u) \in Q_V\}$ in Module 7.

Injective, surjective, inverse. $f : X \to Y$ is *injective* if $f(x) = f(x') \Rightarrow x = x'$, *surjective* if $\forall y\ \exists x : f(x) = y$, and *bijective* if both; then an inverse $f^{-1} : Y \to X$ with $f^{-1}(f(x)) = x$ and $f(f^{-1}(y)) = y$ exists. $2x + 1$ is bijective on $\mathbb{R}$; x^2 is neither on $\mathbb{R}$, but bijective on $[0, \infty)$ with inverse $\sqrt{y}$; $x \mapsto Ax$ is bijective iff A is square and invertible. Strictly increasing functions are injective, which is why Module 5 can invert a continuity modulus ϕ. *Pitfall:* $f^{-1}(B)$ of a *set* is a preimage and always exists; $f^{-1}(y)$ of a *point* needs a bijection.

DEFINITION — SET-VALUED MAP, RELATION, OPERATOR ON SETS

A **set-valued map** $F : X \rightrightarrows Y$ assigns to each x a set $F(x) \subseteq Y$, possibly empty; its graph $\{(x, y) : y \in F(x)\} \subseteq X \times Y$ is a **relation**, and a function is the case where every $F(x)$ has one element. An **operator on sets** $\Phi : 2^X \to 2^X$ maps subsets to subsets, e.g. $\mathrm{Pre}(S) = \{x :\ \exists u \in U,\ f(x, u) \in S\}$, the states from which some input leads into S. A relation $R \subseteq \mathbb{R} \times \mathbb{R}$ is **monotone** if $(a - a')(b - b') \geq 0$ for all $(a, b), (a', b') \in R$.

Examples. Admissible inputs of a safety filter, a GP confidence interval $x \mapsto [\mu_t(x) \pm \beta_t \sigma_t(x)]$, successor sets $\{f(x, u) : u \in U\}$. The sign relation $\{(a, 1) : a > 0\} \cup \{(0, b) : |b| \leq 1\} \cup \{(a, -1) : a < 0\}$ is monotone but, with its vertical segment, not a function. Used in Module 10, Module 12 (activations replaced by $\Delta v \mapsto \{D\Delta v : D \text{ diagonal, entries in } [\alpha, \beta]\}$) and Module 14 (monotone relations).

Fixed points and contractions

DEFINITION — FIXED POINT, FIXED-POINT ITERATION, CONTRACTION

x^* is a **fixed point** of $g : X \to X$ if $g(x^*) = x^*$. **Fixed-point iteration** picks $x_0 \in X$ and sets $x_{k+1} = g(x_k)$. g is a **contraction** with factor $L \in [0, 1)$ if $\|g(x) - g(y)\| \le L\|x - y\|$ for all $x, y \in X$: a Lipschitz map with constant below 1 (Primer B).

Intuition and example. In one dimension fixed points are where the graph of g crosses the diagonal $y = x$; the iteration walks a staircase ("cobweb") between graph and diagonal, which a contraction pulls into the crossing (see the explorer). For $g(x) = 0.5x + 1$, solving $x = 0.5x + 1$ gives $x^* = 2$. From $x_0 = 0$: $x_1 = 1$, $x_2 = 1.5$, $x_3 = 1.75$, $x_4 = 1.875$; the errors $2, 1, 0.5, 0.25, 0.125$ halve because $|g(x) - g(y)| = 0.5|x - y|$. This is value iteration $V_{k+1} = 1 + \gamma V_k$ for one state with reward 1 and $\gamma = 0.5$: the iterates are the partial sums $1 + \gamma + \cdots + \gamma^{k-1}$ of Section 4, with limit $1/(1 - \gamma)$.

THEOREM — BANACH FIXED-POINT THEOREM (CONTRACTION MAPPING THEOREM)

Assumptions. $X \subseteq \mathbb{R}^n$ is nonempty and closed (it contains the limits of its convergent sequences, Section 6); g maps X into X; g is a contraction with factor $L \in [0, 1)$.
Conclusion. g has exactly one fixed point $x^* \in X$, the iterates converge to it from every $x_0 \in X$, and for all k

$$\|x_k - x^*\| \le L^k \|x_0 - x^*\|, \qquad \|x_k - x^*\| \le \frac{L^k}{1 - L} \|x_1 - x_0\|.$$

In words. One fixed point, reached from anywhere at a geometric rate; the a priori bound needs only the first step, so it predicts how many iterations suffice.
Why each assumption matters. $L < 1$: $g(x) = x + 1$ ($L = 1$) has no fixed point, and $g(x) = -x$ ($L = 1$) makes $x_0, -x_0, x_0, \ldots$ oscillate. Self-map: $g(x) = x/2 + 1$ contracts on $[0, 1]$ but its fixed point 2 lies outside. Closedness: $g(x) = x/2$ maps $(0, 1]$ into itself, yet the iterates tend to $0 \notin (0, 1]$. It holds in every complete normed space (Primer A), e.g. for value functions.

Proof of uniqueness and of the first bound. If $g(x^*) = x^*$ and $g(y^*) = y^*$, then $\|x^* - y^*\| = \|g(x^*) - g(y^*)\| \le L\|x^* - y^*\|$, so $(1 - L)\|x^* - y^*\| \le 0$ and $x^* = y^*$. Also $\|x_k - x^*\| = \|g(x_{k-1}) - g(x^*)\| \le L\|x_{k-1} - x^*\|$, and k repetitions give $L^k\|x_0 - x^*\|$. ■ Existence and the a priori bound need limits and the geometric series: walkthrough in Section 4.

WHERE THIS IS USED

Value iteration as a γ-contraction (Primer E); discounted safety Bellman equations with a unique fixed point in Module 10 and Module 9; the constraint-cost evaluation operator in Module 11; equilibrium layers $z = \sigma(Wz + Ux + b)$ in Primer E and Module 13. If the closed-loop map F is a contraction, two trajectories satisfy $\|x_t - y_t\| \le L^t \|x_0 - y_0\|$: this *incremental* stability (contraction) is certified in Module 2, Module 11 and Module 14.

PITFALL — CONVERGENCE IS NOT CONTRACTION

$g(x) = x/(1+x^2)$ sends every start to 0 ($|g(x)| < |x|$ for $x \neq 0$), but $g'(0) = 1$, so no $L < 1$ works near 0, and convergence is slow: from $x_0 = 1$, $x_{10} \approx 0.209$, $x_{100} \approx 0.0700$, $x_{1000} \approx 0.0223$, close to $1/\sqrt{2k}$. Under the strict-decrease, continuity and compactness hypotheses of Section 6, a Lyapunov function ($V = x^2$ here) proves convergence to the equilibrium; it gives no uniform geometric estimate $\|x_t - y_t\| \le q^t \|x_0 - y_0\|$ with $q < 1$ between two trajectories, and none exists here.

Monotone set operators and their limits

FACT — MONOTONE ITERATION ON A FINITE SET

An operator $\Phi : 2^D \to 2^D$ is **monotone** if $A \subseteq B \Rightarrow \Phi(A) \subseteq \Phi(B)$; S is a fixed point if $\Phi(S) = S$. Let D be finite and Φ monotone, and iterate $S_{k+1} = \Phi(S_k)$.
(a) If $S_0 \subseteq \Phi(S_0)$, then $S_0 \subseteq S_1 \subseteq \cdots$ with at most $|D| - |S_0|$ strict increases, and the final set $\bar{S}$ is the **least fixed point** containing S_0.
(b) If $\Phi(X^0) \subseteq X^0$, then $X^0 \supseteq X^1 \supseteq \cdots$ with at most $|X^0|$ strict decreases, and the final set is the **greatest fixed point** contained in X^0.

Proof of (a). If $S_{k-1} \subseteq S_k$, monotonicity gives $S_k = \Phi(S_{k-1}) \subseteq \Phi(S_k) = S_{k+1}$, so by induction the sets grow. Each strict increase adds one of the $|D| - |S_0|$ points outside S_0. Once $S_{k+1} = S_k$, all later sets are equal and $\Phi(\bar{S}) = \bar{S}$. If $T \supseteq S_0$ is a fixed point, $S_k \subseteq T$ implies $S_{k+1} = \Phi(S_k) \subseteq \Phi(T) = T$, so $\bar{S} \subseteq T$. (b) reverses the inclusions. ■

Worked example (SafeOpt's reachable set, Module 4). Grid $D = \{0, \dots, 8\}$, distance $|x - x'|$, Lipschitz constant $L = 1$, threshold $h = 0$ (safe means $f(x) \ge 0$), no estimation error. The operator $R(S) = S \cup \{x \in D : \exists x' \in S : \ f(x') - L|x - x'| \ge h\}$ certifies x through a witness whose value exceeds the threshold by L times the distance; it is monotone and $S \subseteq R(S)$

. With $f = (1.8,\ 2.3,\ 1.4,\ 2.4,\ 1.4,\ 2.4,\ 1.4,\ 0.4,\ 1.4)$ (consecutive values differ by at most 1, so f is 1-Lipschitz) and $S_0 = \{0\}$:

k	S_k	new points (witness check)
1	$\{0, 1\}$	1 from $x' = 0$: $1.8 - 1 \geq 0$
2	$\{0, \ldots, 3\}$	$2, 3$ from $x' = 1$: $2.3 - 2 \geq 0$
3	$\{0, \ldots, 5\}$	$4, 5$ from $x' = 3$: $2.4 - 2 \geq 0$
4	$\{0, \ldots, 7\}$	$6, 7$ from $x' = 5$: $2.4 - 2 \geq 0$
5	$\{0, \ldots, 7\}$	none: the fixed point $\bar{R}(S_0)$

Four strict increases, well within $|D| - |S_0| = 8$. Point 8 is safe ($f(8) = 1.4$) but unreachable: every witness fails, e.g. $0.4 - 1 < 0$ from 7 and $2.4 - 3 < 0$ from 5. This is remark (ii) of Module 4. **Shrinking iterations** compute viability kernels: $X^0 = X \setminus X_F$, $X^{k+1} = X^k \cap \mathrm{Pre}(X^k)$ keeps the states from which some input stays in the current set; by (b) it stops at the largest $Y \subseteq X \setminus X_F$ with $Y \subseteq \mathrm{Pre}(Y)$. Game and shield papers write greatest fixed points as $\nu X.(\cdots)$, least ones as $\mu X.(\cdots)$. Reachability in a directed graph is the least fixed point of $X \mapsto S \cup \mathrm{Post}(X)$, found by breadth-first search (Module 6 adds Dijkstra's shortest paths). *Caution:* SafeOpt's "closure" is this fixed point, not a topological closure, and on continuous domains the iteration need not stop.

> WHERE THIS IS USED
>
> SafeOpt's closure $\bar{R}_\epsilon(S_0)$, "reached after at most $|D|$ steps", in Module 4 and its per-constraint variants in Module 6; reachability and return operators in Module 6; the viability iteration in Module 7; greatest fixed points $\nu X.\, F^g \cap \mathrm{CPre}(X)$ for safety games and shields in Module 7 and Module 10.

Practice: Functions, Maps & Fixed Points

The Easy exercises refresh function notation and substitution. Medium adds preimages and contraction estimates; Hard asks you to explain which assumptions make an argument work. If a definition feels unfamiliar, revisit this section before trying again.

▼ Exercise 0.F1 — Easy: Domain, codomain and actual outputs

Let $f : X \to Y$ be $f(x) = x^2$, where $X = \{-2, -1, 0, 1, 2\}$ and $Y = \{0, 1, 4, 9\}$. State the domain, codomain and image $f(X)$. Is f injective? Is it surjective onto Y?

Review: functions and fixed points

▼ Show hint

The codomain is the declared target set. The image contains only outputs actually reached. Compare $f(-1)$ and $f(1)$.

▼ Show answer

> The domain is X and the codomain is Y. Evaluating the five inputs gives $4, 1, 0, 1, 4$, so the image is $f(X) = \{0, 1, 4\}$.
>
> The map is not injective: distinct inputs -1 and 1 both produce 1. It is not surjective onto Y: the declared output 9 has no preimage in X. Being allowed as an output is different from actually being reached.

▼ Exercise 0.F2 — Easy: Compute two compositions

On $\mathbb{R}$, let $f(x) = 2x - 1$ and $g(x) = x^2$. Find $(f \circ g)(2)$ and $(g \circ f)(2)$, then write a formula for each composition at a general x.

Review: functions and fixed points

▼ Show hint

Apply the function on the right first: $(f \circ g)(x) = f(g(x))$.

▼ Show answer

> First $g(2) = 4$, then $f(4) = 7$, so $(f \circ g)(2) = 7$. In the other order, $f(2) = 3$, then $g(3) = 9$, so $(g \circ f)(2) = 9$.
>
> $$\begin{aligned} (f \circ g)(x) &= 2x^2 - 1, \\ (g \circ f)(x) &= (2x - 1)^2 \\ &= 4x^2 - 4x + 1. \end{aligned}$$
>
> The order of composition matters; the two functions are different.

▼ **Exercise 0.F3 — Easy: A fixed point and three iterations**

Let $g(x) = (x+3)/4$. Solve $g(x^*) = x^*$. Starting at $x_0 = -1$, calculate x_1, x_2, x_3 using $x_{k+1} = g(x_k)$. What happens to the distance from the fixed point at each step?

Review: functions and fixed points

▼ **Show hint**

Multiply the fixed-point equation by 4. For the iteration, feed each new output back into g.

▼ **Show answer**

$x^* = (x^*+3)/4$ gives $4x^* = x^* + 3$, hence $x^* = 1$.

$$\begin{aligned} x_1 &= (-1+3)/4 = 1/2, \\ x_2 &= (1/2+3)/4 = 7/8, \\ x_3 &= (7/8+3)/4 = 31/32. \end{aligned}$$

The distances from 1 are $2, 1/2, 1/8, 1/32$ for $k = 0, 1, 2, 3$. Each is one quarter of the previous distance, since $g(x) - 1 = (x-1)/4$.

▼ **Exercise 0.F4 — Medium: Images and preimages of intervals**

For $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, find $f([-3, 1])$, $f^{-1}([1, 4])$, and $f^{-1}(\{-1\})$. Explain why the preimage notation is meaningful even though this f has no inverse function.

Review: functions and fixed points

▼ **Show hint**

For an image, collect squares of permitted inputs. For a preimage, solve the output condition for every possible real input, including negative inputs.

▼ **Show answer**

> $f([-3, 1]) = [0, 9]$: 0 is reached at $x = 0$, 9 at $x = -3$, and every value $y \in [0, 9]$ is reached by $x = -\sqrt{y} \in [-3, 0]$.
>
> For the second set, $1 \le x^2 \le 4$ means $1 \le |x| \le 2$, so $f^{-1}([1, 4]) = [-2, -1] \cup [1, 2]$. No real square is -1, so $f^{-1}(\{-1\}) = \emptyset$.
>
> A preimage is a set of all inputs passing an output test. It does not require a unique input for each output. An inverse function would require that uniqueness and surjectivity; x^2 on $\mathbb{R}$ has neither.

▼ Exercise 0.F5 — Medium: The same formula with four choices of domain and codomain

Classify each map as injective, surjective, both, or neither. All use the formula $f(x) = x^2$.

1. $\mathbb{R} \to \mathbb{R}$.
2. $\mathbb{R} \to [0, \infty)$.
3. $[0, \infty) \to \mathbb{R}$.
4. $[0, \infty) \to [0, \infty)$.

For the bijective map, write its inverse.

Review: functions and fixed points

▼ Show hint

Restricting inputs can remove repeated outputs. Restricting the declared target can remove unreachable outputs.

▼ Show answer

1. Neither: $f(-1) = f(1)$ prevents injectivity, and no negative output is reached.
2. Surjective only: every $y \geq 0$ is reached by $x = \sqrt{y}$, but the repeated inputs $-1, 1$ remain.
3. Injective only: for nonnegative x, x', $x^2 = (x')^2$ implies $x = x'$. Negative outputs in the codomain are still unreachable.
4. Both: the two arguments above prove injectivity and surjectivity. The inverse is $f^{-1}(y) = \sqrt{y}$ on $[0, \infty)$.

Domain and codomain are part of a function's specification; its formula alone does not determine these properties.

▼ Exercise 0.F6 — Medium: An alternating contraction and an error target

Let $g(x) = -x/2 + 3$ on $\mathbb{R}$, and $x_0 = 0$. Find the fixed point and the first three iterates. Prove that g has contraction factor $1/2$. Use the exact error to find the smallest integer $k \geq 0$ for which $|x_k - x^*| \leq 1/100$.

Review: functions and fixed points

▼ Show hint

Subtract the fixed-point equation from the iteration. The sign alternates, but the absolute error halves.

▼ Show answer

Solving $x^* = -x^*/2 + 3$ gives $x^* = 2$. The iterates are $x_1 = 3, x_2 = 3/2, x_3 = 9/4$. Also $|g(x) - g(y)| = |x - y|/2$, proving the contraction property.

$$\begin{aligned} x_{k+1} - 2 &= -\tfrac{1}{2}(x_k - 2), \\ x_k - 2 &= -2(-\tfrac{1}{2})^k, \\ |x_k - 2| &= 2(1/2)^k. \end{aligned}$$

At $k = 7$ the error is $1/64 > 1/100$; at $k = 8$ it is $1/128 < 1/100$. Errors decrease each step, so $k = 8$ is the first qualifying iteration. A negative slope causes alternation without preventing contraction.

▼ Exercise 0.F7 — Hard: Why images do not preserve intersections

For $f(x) = x^2$, $A = \{-2\}$ and $B = \{2\}$, compare $f(A \cap B)$ with $f(A) \cap f(B)$. Then prove that for any injective function $f : X \to Y$ and any $A, B \subseteq X$, these two sets are equal.

Review: functions and fixed points

▼ Show hint

For the general proof, show both inclusions. If an output comes from $a \in A$ and $b \in B$, injectivity forces $a = b$.

▼ Show answer

> Here $A \cap B = \emptyset$, so $f(A \cap B) = \emptyset$. But $f(A) = f(B) = \{4\}$, making $f(A) \cap f(B) = \{4\}$. Different inputs produced the same output.
>
> For any function, if $y \in f(A \cap B)$, there is an $x \in A \cap B$ with $f(x) = y$. Thus y belongs to both $f(A)$ and $f(B)$; this proves one inclusion.
>
> Conversely, suppose $y \in f(A) \cap f(B)$. There are $a \in A$ and $b \in B$ with $f(a) = y = f(b)$. When f is injective, $a = b$. This common input belongs to $A \cap B$, so $y \in f(A \cap B)$. The reverse inclusion follows, proving equality.

▼ Exercise 0.F8 — Hard: Grow a set until it stops

Let $D = \{0, 1, 2, 3, 4\}$, with $F(0) = 1, F(1) = 2, F(2) = 2, F(3) = 4, F(4) = 3$. Define $\Phi(S) = S \cup F(S)$ for $S \subseteq D$, where $F(S) = \{F(x) : x \in S\}$. Start with $S_0 = \{0\}$ and iterate $S_{k+1} = \Phi(S_k)$. Find the final set, prove Φ is monotone, and explain why $3, 4$ are not added.

Review: functions and fixed points

▼ Show hint

Use only the states already in S_k to compute $F(S_k)$. For monotonicity, follow an arbitrary element from $A \subseteq B$.

▼ Show answer

$$
\begin{aligned}
S_1 &= \{0\} \cup \{1\} = \{0, 1\}, \\
S_2 &= \{0, 1\} \cup \{1, 2\} = \{0, 1, 2\}, \\
S_3 &= \{0, 1, 2\} \cup \{1, 2\} = S_2.
\end{aligned}
$$

Hence S_2 is a fixed point of Φ, and all later iterates equal it. There are two strict increases.

If $A \subseteq B$, every input used to form $F(A)$ also belongs to B, so $F(A) \subseteq F(B)$. Combining this with $A \subseteq B$ gives $A \cup F(A) \subseteq B \cup F(B)$, proving monotonicity. States $3, 4$ form a separate cycle: neither is the successor of $0, 1, 2$, so neither can be reached from the seed 0.

▼ **Exercise 0.F9 — Hard: Audit the contraction theorem**

In each case, say which assumption of the contraction theorem fails and whether a fixed point in X exists.

1. $X = (0, 1]$, $g(x) = x/2$.
2. $X = [0, 1]$, $g(x) = x/2 + 1$.
3. $X = [-1, 1]$, $g(x) = -x$.

In (3), describe the iteration from $x_0 = 1$.

Review: functions and fixed points

▼ **Show hint**

Check closedness, whether every output stays in X, and whether a single factor $L < 1$ bounds all distances. Solve $g(x) = x$ separately.

▼ **Show answer**

1. Closedness fails: $1, 1/2, 1/4, \ldots$ lie in X and tend to $0 \notin X$. The map stays in X and contracts by $1/2$, but its only possible fixed point is 0, so it has none in X.
2. The self-map assumption fails: $g(1) = 3/2 \notin X$. It contracts by $1/2$ and X is closed, but its fixed-point equation gives $x = 2 \notin X$.
3. Strict contraction fails: $|g(x) - g(y)| = |x - y|$, so a factor below 1 cannot work for distinct inputs. The fixed point 0 exists, but from 1 the iterates are $1, -1, 1, -1, \ldots$ and do not converge.

A failed assumption removes the theorem's guarantee. It does not by itself prove that every part of the conclusion is false.

3. Reading and Writing Proofs

A proof is a chain of statements, each following from assumptions, definitions or earlier steps by a rule you can name.

FACT — A READING CHECKLIST FOR THEOREM BOXES

(1) Separate assumptions from conclusion and list every quantifier: which set, all t at once or each t, probability $1 - \delta$ per run or per round? (2) Check the conclusion on the simplest instance. (3) For each assumption, find what breaks without it (the modules' "Why each assumption matters"). (4) While reading the proof, mark where each assumption is used; an unused one is redundant or signals a misreading.

Step 4 pays off in Module 5: the LoSBO safety proof uses only the Lipschitz constant, the noise bound and the seed, never β_t, the kernel or the RKHS norm, hence "for any β_t".

Direct proofs and chains of inequalities

LEMMA — THE LIPSCHITZ CERTIFICATE BEHIND SAFEOPT AND LOSBO

If $|f(x) - f(x')| \le L|x - x'|$ for all x, x' and $f(x') - L|x - x'| \ge h$, then $f(x) \ge h$.

Proof (a direct proof: a chain whose links each have one reason and point the same way):

$$f(x) \overset{(1)}{=} f(x') + \big(f(x) - f(x')\big) \overset{(2)}{\ge} f(x') - L|x - x'| \overset{(3)}{\ge} h.$$

(1) adds and subtracts $f(x')$; (2) uses $f(x) - f(x') \ge -|f(x) - f(x')| \ge -L|x - x'|$, only one side of the absolute value; (3) is the hypothesis. With $f(x') = 2.5$, $L = 1$, $|x - x'| = 1.5$, $h = 0$: $f(x) \ge 1 \ge 0$. ■

Tools for chains. $\|a + b\| \le \|a\| + \|b\|$ and $\big|\|a\| - \|b\|\big| \le \|a - b\|$; multiplying by $c \ge 0$ keeps the direction, by $c < 0$ flips it; increasing functions (e^x, $\ln x$, $\sqrt{x}$) keep it. One strict link makes the chain strict.

Contrapositive, contradiction, pigeonhole

To prove $P \Rightarrow Q$ one may prove the **contrapositive** $\neg Q \Rightarrow \neg P$. Example: "if $\|x\| \le \varepsilon$ for every $\varepsilon > 0$, then $x = 0$"; if $x \neq 0$, then $\varepsilon = \|x\|/2$ violates the premise. ■ ("To show $a = b$, show $|a - b| \le \varepsilon$ for every $\varepsilon > 0$" returns in every limit argument.) A **proof by contradiction** assumes the conclusion false and derives something impossible. The **pigeonhole principle** is the simplest case: $n + 1$ objects in n boxes put two in one box.

FACT — DETERMINISTIC LOOPS ON FINITE STATE SPACES CYCLE

If X is finite and $x_{t+1} = F(x_t)$ for a fixed $F : X \to X$ (stationary deterministic feedback and dynamics), every state the trajectory ever visits appears among $x_0, \dots, x_{|X|-1}$; so if it ever enters a failure set, it does so by time $|X| - 1$.

Proof. The $|X| + 1$ states $x_0, \dots, x_{|X|}$ lie in a set of size $|X|$, so $x_i = x_j$ for some $0 \le i < j \le |X|$ (pigeonhole). Determinism gives $x_{i+m} = x_{j+m}$ for all $m \ge 0$ (induction on m), so from time i on the trajectory repeats $x_i, \dots, x_{j-1}$, and $j - 1 \le |X| - 1$. ■

Induction, including induction over time

> **DEFINITION — PRINCIPLE OF MATHEMATICAL INDUCTION**
>
> If $P(0)$ is true (**base case**) and $P(t) \Rightarrow P(t+1)$ for every $t \in \mathbb{N}$ (**induction step**), then $P(t)$ holds for all $t \in \mathbb{N}$. In **strong induction** the step may use all of $P(0), \ldots, P(t)$.

Induction over time steps proves "safe at every step": safe now, and safe now plus the certificate imply safe next. The walkthrough proves a discrete-time barrier certificate this way and shows the trap to avoid.

DERIVATION: A SAFETY PROOF BY INDUCTION OVER TIME STEPS

1. Claim | 2. P(t), base | 3. Step | 4. Audit | 5. Trap

Step 1: The claim and its assumption

A closed loop $x_{t+1} = F(x_t)$ and a safe set $C = \{x : h(x) \geq 0\}$. Assumption (A): for some $\alpha \in [0, 1]$, $h(F(x)) \geq (1 - \alpha)\, h(x)$ for every $x \in C$, so from inside C the barrier value shrinks by at most the factor $1 - \alpha$ per step. Claim: $x_0 \in C$ implies $x_t \in C$ for all $t \in \mathbb{N}$. Example: $F(x) = x^2$ and $h(x) = 1 - x^2$, so $C = [-1, 1]$.

$$h(F(x)) = 1 - x^4 = (1 - x^2)(1 + x^2) = h(x)\,(1 + x^2) \;\geq\; h(x) \qquad \text{whenever } h(x$$

> **KEY INSIGHT**
>
> So (A) holds with $\alpha = 0$, but only on C: where $h(x) < 0$ the factor $1 + x^2 > 1$ makes $h(F(x)) < h(x)$. An assumption that holds on part of the space may only be used there.

← Previous Next →

Counterexamples and "it suffices to show"

One counterexample disproves a "for all". "Every bounded sequence converges": $(-1)^t$ does not. "A continuous function on a bounded set attains its maximum": x on $(0, 1)$ does not. "If V decreases along trajectories, they converge to 0": Module 11 gives a discontinuous F with $x_t = 1 + 2^{-t} \to 1$. "A barrier condition on the boundary suffices": it can fail where $\nabla h = 0$ (Module 1; Section 6 below). "**It suffices to show** Q" means that Q implies the goal: to show $A = B$ it suffices to show

$A \subseteq B$ and $B \subseteq A$; LoSBO's proof starts "it suffices that $f \geq h$ on every S_t", because every query is taken from S_t.

> **WHERE THIS IS USED**
>
> Induction over rounds and the assumption audit in Module 5; "induction removes the circularity" in Module 14; "by induction, X^k is the set of states from which failure can be avoided for k steps" and finite-state cycles in Module 7; recursive feasibility of predictive safety filters in Module 10.

> **COMMON MISTAKES IN PROOFS**
>
> (1) *Circular reasoning*: assuming the trajectory stays where a certificate holds in order to prove that it stays there. (2) *Checking an example instead of the claim*: one safe simulation proves nothing about all initial states. (3) *Using an assumption outside its domain*, e.g. a sector bound valid on a box, applied outside the box. (4) *Multiplying by a quantity of unknown sign*.

Practice: Reading and Writing Proofs

Write a reason beside each step. Easy exercises practise one proof move; Medium builds complete arguments; Hard combines the proof with an audit of its assumptions. If a definition feels unfamiliar, revisit this section before trying again.

▼ Exercise 0.P1 — Easy: A direct proof by completing the square

Prove $x^2 + 2x + 2 \geq 1$ for every real x. When does equality hold? Give a general argument, not just numerical checks.

Review: proof techniques

▼ Show hint

Rewrite $x^2 + 2x + 2$ as one square plus a constant.

▼ Show answer

$$x^2 + 2x + 2 = (x + 1)^2 + 1 \geq 1.$$

The equality is obtained by expanding the square; the inequality uses $(x + 1)^2 \geq 0$ for every real x. Equality holds exactly when $x + 1 = 0$, hence when $x = -1$. Because x was arbitrary, this proves the claim for all real inputs.

▼ Exercise 0.P2 — Easy: Prove a contrapositive about even numbers

Let n be an integer. Prove: if n^2 is even, then n is even. Use the contrapositive. Recall that an even integer has the form $2m$ and an odd integer the form $2m + 1$, for an integer m.

Review: proof techniques

▼ Show hint

The contrapositive says: if n is odd, then n^2 is odd. Substitute $n = 2m + 1$ and expand.

▼ Show answer

Assume n is odd, so $n = 2m + 1$ for some integer m. Then

$$\begin{aligned} n^2 &= (2m + 1)^2 \\ &= 4m^2 + 4m + 1 \\ &= 2(2m^2 + 2m) + 1. \end{aligned}$$

The number $2m^2 + 2m$ is an integer, so the last expression is odd. This proves "n odd implies n^2 odd", the contrapositive of the requested claim. Every integer is even or odd, so these are exactly the required negations.

▼ Exercise 0.P3 — Easy: Disprove three universal claims

Give a counterexample to each claim over real numbers, and explicitly check why it fails.

1. For every x, $x^2 \geq x$.
2. If $xy = 0$, then both $x = 0$ and $y = 0$.
3. If $x^2 = y^2$, then $x = y$.

Review: proof techniques

▼ Show hint

Try a number between 0 and 1 for (1), one zero factor for (2), and opposite nonzero numbers for (3).

▼ Show answer

1. Take $x = 1/2$. Its square is $1/4 < 1/2$, contradicting the proposed inequality.
2. Take $x = 0, y = 1$. Their product is 0, but $y \neq 0$, so the premise holds and the conclusion fails.
3. Take $x = -1, y = 1$. Both squares are 1, but the numbers differ.

One valid counterexample is enough to refute a universal claim. Many examples where it works would not prove it.

▼ Exercise 0.P4 — Medium: Write an induction proof from start to finish

Prove that $\sum_{j=1}^{n} j = n(n+1)/2$ for every integer $n \geq 0$, with the sum at $n = 0$ defined to be 0. State the base case, induction hypothesis, induction step and conclusion explicitly.

Review: proof techniques

▼ Show hint

For the step, split the sum at $n+1$ into the old sum and the new term $n+1$.

▼ Show answer

Base case: at $n = 0$, the empty sum is $0 = 0(0+1)/2$.

Induction hypothesis: assume for some $n \geq 0$ that the sum through n is $n(n+1)/2$.

Induction step:

$$\begin{aligned}\sum_{j=1}^{n+1} j &= \sum_{j=1}^{n} j + (n+1) \\ &= \frac{n(n+1)}{2} + (n+1) \\ &= \frac{(n+1)(n+2)}{2}.\end{aligned}$$

The first line separates the final term, the second uses the hypothesis, and the third factors out $n+1$. This is precisely the claimed formula with $n+1$ in place of n.

Conclusion: by induction, the formula holds for every integer $n \geq 0$.

▼ Exercise 0.P5 — Medium: Repair a proof with a sign mistake

A proposed proof says: "If $a \le b$, multiplying by a gives $a^2 \le ab$, and multiplying by b gives $ab \le b^2$. Hence $a^2 \le b^2$." Explain the gap, give a counterexample, and repair the argument under the additional assumption $0 \le a \le b$.

Review: proof techniques

▼ Show hint

Multiplication by a negative number reverses an inequality. Alternatively factor $b^2 - a^2$.

▼ Show answer

> The proof never checks the signs of a and b. Its multiplication steps preserve the inequality only when the multiplier is nonnegative. For $a = -3, b = -2$, the premise $-3 \le -2$ is true but the conclusion $9 \le 4$ is false.
>
> Under $0 \le a \le b$, both multipliers are nonnegative, so the original two steps are valid. Equivalently, $b^2 - a^2 = (b - a)(b + a) \ge 0$: both factors are nonnegative. Thus $a^2 \le b^2$ follows with the added assumption.

▼ Exercise 0.P6 — Medium: Prove that an interval stays invariant

Let $x_{t+1} = x_t/2 + 1/4$ and $x_0 \in C = [0, 1]$. First show that any point of C maps back into C. Then use induction to prove $x_t \in C$ for all integers $t \ge 0$. Where is the assumption on x_0 used?

Review: proof techniques

▼ Show hint

If $0 \le x \le 1$, bound $x/2 + 1/4$ from below and above. Use membership in C as the induction statement.

▼ Show answer

> If $x \in C$, then $0 \le x \le 1$, so $1/4 \le x/2 + 1/4 \le 3/4$. Since $[1/4, 3/4] \subseteq [0, 1]$, the next state belongs to C.
>
> For induction, let $P(t)$ mean $x_t \in C$. The base case $P(0)$ is exactly the assumption $x_0 \in C$. If $P(t)$ holds, apply the one-step argument at $x = x_t$ to obtain $P(t + 1)$. Thus $P(t)$ holds at every time.
>
> Without the initial-state assumption the base case can fail: $x_0 = 2$ lies outside C, and even $x_1 = 5/4$ is outside.

▼ Exercise 0.P7 — Hard: Prove uniqueness without assuming existence

Let X be a nonempty subset of $\mathbb{R}$ and $g : X \to X$ satisfy $|g(x) - g(y)| \leq |x - y|/3$ for all $x, y \in X$. Prove that g has at most one fixed point. Does your proof also show that a fixed point exists? Give a counterexample if it does not.

Review: proof techniques

▼ Show hint

Assume two fixed points a, b and apply the contraction inequality to them. For existence, consider an interval missing its limit point.

▼ Show answer

Suppose $g(a) = a$ and $g(b) = b$. Write $d = |a - b| \geq 0$. Then

$$d = |g(a) - g(b)| \leq d/3.$$

Subtracting $d/3$ gives $2d/3 \leq 0$. Together with $d \geq 0$ this forces $d = 0$, hence $a = b$.

This proves only that two different fixed points cannot exist. For a counterexample to existence, take $X = (0, 1]$ and $g(x) = x/3$. It maps X into itself and has the stated contraction factor, but $x = x/3$ forces $x = 0 \notin X$. Closedness, an extra hypothesis of the full theorem, is missing.

▼ **Exercise 0.P8 — Hard: Remove circular reasoning from a safety proof**

Let $C = [-1, 1]$ and suppose a map $F : \mathbb{R} \to \mathbb{R}$ satisfies $|F(x)| \leq (4/5)|x|$ only for $x \in C$. From $x_0 \in C$, define $x_{t+1} = F(x_t)$. A proposed proof says: "The trajectory is in C, so the bound applies at every step, so the trajectory stays in C." Explain the circularity and replace it with an induction proof of $|x_t| \leq (4/5)^t|x_0|$ and $x_t \in C$.

Review: proof techniques

▼ **Show hint**

At time t, use the bound already proved at that time to establish membership in C before applying the hypothesis to $F(x_t)$.

▼ **Show answer**

The proposed proof assumes the all-time membership that it is meant to establish. The hypothesis on F cannot be used outside C.

At $t = 0$, the quantitative bound is equality, and $|x_0| \leq 1$ by assumption. For the step, assume $|x_t| \leq (4/5)^t|x_0|$. Since $(4/5)^t \leq 1$, this implies $|x_t| \leq 1$, so $x_t \in C$ and the bound on F is available. Therefore

$$\begin{aligned} |x_{t+1}| &= |F(x_t)| \\ &\leq (4/5)|x_t| \\ &\leq (4/5)^{t+1}|x_0| \\ &\leq 1. \end{aligned}$$

This proves the quantitative bound at $t + 1$ and membership in C there. Induction proves both claims at every time, without assuming future membership.

▼ **Exercise 0.P9 — Hard: When a finite simulation proves an all-time claim**

A fixed deterministic map on $X = \{a, b, c, d\}$ has $F(a) = b$, $F(b) = c$, $F(c) = b$, $F(d) = d$. Start at $x_0 = a$; state d is failure. List the first five states. Explain why d is never reached. More generally, justify why checking $x_0, \ldots, x_3$ suffices for any fixed deterministic map on this four-state set. Can that conclusion fail if the map changes with time?

Review: proof techniques

▼ **Show hint**

Among five states chosen from a set of four, two must repeat. A fixed deterministic map forces the same future after a repeated state.

▼ **Show answer**

> The first five states are a, b, c, b, c. Once b repeats, its successor is again c, whose successor is b. Thus the trajectory cycles through b, c and never visits d.
>
> For any fixed map on four states, among $x_0, \ldots, x_4$ there are indices $0 \le i < j \le 4$ with $x_i = x_j$. Applying the same map gives $x_{i+1} = x_{j+1}$; repeating this argument gives $x_{i+m} = x_{j+m}$ for every $m \ge 0$. The trajectory from i therefore cycles through $x_i, \ldots, x_{j-1}$. Every state ever visited already appeared by time $j - 1 \le 3$, proving the general claim.
>
> The conclusion can fail with a changing map: let F_t send every state to a for $t = 0, 1, 2$ and to d for $t \ge 3$. Starting at a, the first four states are all a, but $x_4 = d$. Repetition alone is insufficient when the future rule changes.

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4. Sequences, Limits & Series

Iterates, trajectories and learning curves are sequences; most guarantees concern their limits and rates.

DEFINITION — CONVERGENCE, CAUCHY SEQUENCES

A sequence $(a_t)_{t\in\mathbb{N}}$ **converges** to a, written $a_t \to a$ or $\lim_{t\to\infty} a_t = a$, if

$$\forall \varepsilon > 0 \;\; \exists T \in \mathbb{N} \;\; \forall t \geq T : \quad |a_t - a| < \varepsilon.$$

For vectors use $\|\cdot\|$ (equivalently, every coordinate converges). $a_t \to \infty$ means $\forall M \; \exists T \; \forall t \geq T : a_t > M$. A **Cauchy sequence** satisfies $\forall \varepsilon > 0 \; \exists T \; \forall s, t \geq T : |a_s - a_t| < \varepsilon$; in $\mathbb{R}^n$ a sequence converges iff it is Cauchy (*completeness*).

Intuition and example. Whatever tolerance ε you name, the sequence is eventually within ε of a and stays there; T may grow as ε shrinks. $a_t = 1/(t+1) \to 0$: any integer $T \geq 1/\varepsilon$ works, since $t \geq T$ gives $1/(t+1) \leq 1/(T+1) < 1/T \leq \varepsilon$ ($T = 100$ for $\varepsilon = 0.01$). $(-1)^t$ has no limit: with $\varepsilon = 1$, both 1 and -1 would lie within 1 of a, and then $2 \leq |1 - a| + |a + 1| < 2$.

Rules. Limits respect sums, products and quotients (nonzero denominator limit). $a_t \leq b_t$ for all t gives $\lim a_t \leq \lim b_t$, but strict inequalities may become equalities ($1/(t+1) > 0$, limit 0). **Squeeze:** $a_t \leq c_t \leq b_t$ with $a_t, b_t \to \ell$ gives $c_t \to \ell$. **Continuity:** if g is continuous at a and $a_t \to a$, then $g(a_t) \to g(a)$.

THEOREM — MONOTONE CONVERGENCE FOR SEQUENCES

A nonincreasing sequence that is bounded below converges, to $\inf_t a_t$ (Section 5); a nondecreasing one bounded above converges to $\sup_t a_t$. *Proof.* Let $a = \inf_t a_t$. For $\varepsilon > 0$, $a + \varepsilon$ is not a lower bound, so $a_T < a + \varepsilon$ for some T, and for $t \geq T$, $a \leq a_t \leq a_T < a + \varepsilon$. ■ Both assumptions matter: $-t$ decreases without bound, and $(-1)^t$ is bounded but not monotone.

Example (Module 11). If $V \geq 0$ and $V(x_{t+1}) \leq V(x_t)$, then $V(x_t) \to v^* \geq 0$, but nothing forces $v^* = 0$ ($1 + \frac{1}{t+1}$ decreases to 1); that needs compactness and strict decrease (end of Section 6). For $x_{t+1} = x_t/(1 + x_t^2)$, $V(x) = x^2$, $x_0 = 1$, the values (rounded where needed) $1,\ 0.25,\ 0.16,\ 0.119,\ 0.095, \ldots$ decrease and converge. **Pointwise vs uniform:** $f_n \to f$ pointwise if $f_n(x) \to f(x)$ for each x (T may depend on x), uniformly if $\sup_x |f_n(x) - f(x)| \to 0$; $x^n \to 0$ pointwise on $[0, 1)$ but not uniformly. On a finite set pointwise implies uniform (take the largest T).

WHERE THIS IS USED

"$V(x_t)$ decreases to some $v^* \geq 0$" in Module 11; the decreasing sequence $L \geq T[L] \geq T^2[L] \geq \cdots$ in Module 11; monotone confidence endpoints $l_t \uparrow$, $u_t \downarrow$ in Module 4; "nonincreasing and bounded below, so it converges" on finitely many state-action pairs in Module 7.

Series, the geometric series and discounting

THEOREM — GEOMETRIC SERIES

A series $\sum_{t=0}^{\infty} a_t$ is the limit of its partial sums $S_T = \sum_{t=0}^{T-1} a_t$ (T terms), if it exists. For $\gamma \neq 1$, and for $|\gamma| < 1$ respectively,

$$\sum_{t=0}^{T-1} \gamma^t = \frac{1-\gamma^T}{1-\gamma}, \qquad \sum_{t=0}^{\infty} \gamma^t = \frac{1}{1-\gamma}, \qquad \sum_{t=T}^{\infty} \gamma^t = \frac{\gamma^T}{1-\gamma}.$$

Proof. $S_T - \gamma S_T = (1+\gamma+\cdots+\gamma^{T-1}) - (\gamma+\cdots+\gamma^T) = 1-\gamma^T$; divide by $1-\gamma$. If $|\gamma| < 1$, then $|\gamma|^T \to 0$, so $S_T \to \frac{1}{1-\gamma}$, and the tail is $\frac{1}{1-\gamma} - S_T = \frac{\gamma^T}{1-\gamma}$. ■

Discounting. If $|r_t| \leq R_{\max}$ and $0 \leq \gamma < 1$, then $\left|\sum_t \gamma^t r_t\right| \leq \frac{R_{\max}}{1-\gamma}$ and the tail from T on is at most $\frac{\gamma^T R_{\max}}{1-\gamma}$; $1/(1-\gamma)$ is the **effective horizon** ($\gamma = 0.99$ counts about 100 steps).

Example ($\gamma = 0.9$). The limit is 10. With $0.9^{10} \approx 0.3487$, the first ten terms give $S_{10} \approx 6.513$ and the tail is approximately 3.487. Capturing 99% needs $\gamma^T \leq 0.01$, i.e. $T \geq \ln 0.01 / \ln 0.9 \approx 43.7$: 44 terms. Because $\ln(1/\gamma) \geq 1-\gamma$ (proved below), $T \geq \ln(1/\varepsilon)/(1-\gamma)$ always guarantees $\gamma^T \leq \varepsilon$; here it asks for 47. The explorer's lower panel computes this for any γ.

Going deeper — the matrix geometric (Neumann) series ▾

If a square matrix has operator norm $\|A\| < 1$ (Primer A), then $(I-A)^{-1} = \sum_{t \geq 0} A^t$: indeed $(I-A)(I+A+\cdots+A^{T-1}) = I - A^T$ and $\|A^T\| \leq \|A\|^T \to 0$. For an MDP transition matrix P_π (rows sum to 1, so $\|\gamma P_\pi\|_\infty = \gamma$), $(I-\gamma P_\pi)^{-1} = \sum_t \gamma^t P_\pi^t$ collects the discounted visits: the "Neumann series read along trajectories" of Module 9. Module 11 uses $(I-P)^{-1} = \sum_t P^t$ for transient chains.

Telescoping. $\sum_{t=0}^{T-1}(b_{t+1}-b_t)=b_T-b_0$: intermediate terms cancel in pairs. Since $\frac{1}{r(r+1)}=\frac{1}{r}-\frac{1}{r+1}$, $\sum_{r=1}^{R}\frac{1}{r(r+1)}=1-\frac{1}{R+1}\to 1$, which is why a failure probability δ' can be split into pieces $\delta'/(r(r+1))$ (Module 5). Summing a Lyapunov or dissipation inequality $V(x_{t+1})-V(x_t)\le-\varepsilon\|x_t\|^2$ over $t<T$ telescopes the left side, and with $V\ge 0$:

$$\varepsilon\sum_{t=0}^{T-1}\|x_t\|^2\ \le\ V(x_0)-V(x_T)\ \le\ V(x_0)\qquad\text{for every } T.$$

WHERE THIS IS USED

Summed dissipation inequalities in Module 2, Module 12 and Module 13; the performance difference lemma in Module 9; a telescoping $\log\det$ in Module 3; $R_{\max}/(1-\gamma)$ in Module 6, $(1-\gamma^{T_f+1})/(1-\gamma)$ in Module 7 and $\frac{1}{1-\gamma}\langle\rho_\pi,r\rangle$ in Module 8.

FACT — SERIES WITH NONNEGATIVE TERMS

(a) If $a_t\ge 0$, the partial sums are nondecreasing, so $\sum_t a_t$ converges iff they are bounded. (b) **Comparison:** $0\le a_t\le b_t$ and $\sum b_t<\infty$ give $\sum a_t<\infty$; and $\sum|a_t|<\infty$ implies that $\sum a_t$ converges. (c) If $\sum_t a_t$ converges, then $a_t=S_{t+1}-S_t\to s-s=0$. (d) The converse of (c) fails: $\sum_{t\ge1}\frac{1}{t}=\infty$, because the 2^j terms from 2^j+1 to 2^{j+1} are each at least $2^{-(j+1)}$, adding at least $\frac{1}{2}$ per block. In contrast $\sum_{t\ge1}\frac{1}{t^2}\le 1+\sum_{t\ge2}\frac{1}{t(t-1)}=2$ by telescoping.

Application. For $\varepsilon>0$, $\varepsilon\sum_{t<T}\|x_t-x_*\|^2\le V(x_0)$ for all T gives $x_t\to x_*$ by (a) and (c): Step D in Module 14; likewise in Module 11.

Going deeper — when may limits, sums and expectations be exchanged?

A limit and a sum cannot always be exchanged: for $a_{n,t}=1$ if $t=n$ (else 0), $\lim_n\sum_t a_{n,t}=1$ but $\sum_t\lim_n a_{n,t}=0$, although all terms are nonnegative. Two sums, or a sum and an expectation, may be exchanged when the terms are **nonnegative** (Tonelli), e.g. $\mathbb{E}\big[\sum_t X_t\big]=\sum_t\mathbb{E}[X_t]$ for $X_t\ge 0$. A limit may be exchanged with a sum or an expectation under **domination**: $a_{n,t}\to a_t$ with $|a_{n,t}|\le b_t$, $\sum_t b_t<\infty$, gives $\sum_t a_{n,t}\to\sum_t a_t$, and $Y_n\to Y$ almost surely (Section 5) with $|Y_n|\le Z$, $\mathbb{E}[Z]<\infty$, gives $\mathbb{E}[Y_n]\to\mathbb{E}[Y]$ (dominated convergence; bounded convergence if Z is constant). Nonnegative terms that increase with n are another sufficient condition (monotone convergence).

For discounted sums the geometric tail suffices: if $|r_t| \le R_{\max}$, then $\left|\mathbb{E}\big[\sum_t \gamma^t r_t\big] - \sum_{t<T} \gamma^t \mathbb{E}[r_t]\right| \le \frac{\gamma^T R_{\max}}{1-\gamma} \to 0$, so $\mathbb{E}\big[\sum_t \gamma^t r_t\big] = \sum_t \gamma^t \mathbb{E}[r_t]$: "bounded rewards and $\gamma < 1$ kill the tail" in Module 9, the occupancy measures of Module 8. The CVaR walkthrough of Module 1 differentiates inside an expectation by bounded convergence.

Limits of functions; the exponential and the logarithm

$\lim_{x\to a} f(x) = \ell$ means $\forall \varepsilon > 0\ \exists \delta > 0 : 0 < |x-a| < \delta \Rightarrow |f(x) - \ell| < \varepsilon$, equivalently $f(x_t) \to \ell$ for every sequence $x_t \to a$ with $x_t \neq a$; one-sided limits $x \downarrow a$, $x \uparrow a$ use only $x > a$ or $x < a$, and f is continuous at a iff $\lim_{x\to a} f(x) = f(a)$.

KEY EQUATION — THE EXPONENTIAL AS A LIMIT, AND TWO INEQUALITIES

$$e^x = \lim_{n\to\infty} \left(1 + \frac{x}{n}\right)^n, \qquad 1 + x \le e^x \ \ (x \in \mathbb{R}), \qquad \ln x \le x - 1 \ \ (x > 0).$$

$(1+\frac{1}{n})^n \approx 2.5937,\ 2.7048,\ 2.7169$ for $n = 10, 100, 1000$ ($e \approx 2.71828$), and $(1-\frac{1}{n})^n \approx 0.3487,\ 0.3660,\ 0.3677$ ($1/e \approx 0.3679$). *Proof of the inequalities.* $\phi(x) = e^x - 1 - x$ has $\phi'(x) = e^x - 1$, negative for $x < 0$ and positive for $x > 0$, so $\phi \ge \phi(0) = 0$. Replacing x by $x - 1$ gives $x \le e^{x-1}$; take logarithms. ■

Consequences. $(1-p)^n \le e^{-pn}$: n independent trials all miss an event of probability p with probability at most e^{-pn} (for $p = 0.01$, $n = 300$: $0.99^{300} \le e^{-3}$, with $0.99^{300} \approx 0.0490$ and $e^{-3} \approx 0.0498$), the origin of sample sizes like $\ln(1/\beta)/\varepsilon$ in Module 15. With $x = \gamma$: $\ln(1/\gamma) \ge 1 - \gamma$, so $\gamma^T \le e^{-(1-\gamma)T}$. **A soft maximum becomes a maximum:** for $q_1, \dots, q_n$ with maximum m and $\alpha > 0$, $e^{m/\alpha} \le \sum_i e^{q_i/\alpha} \le n\, e^{m/\alpha}$, so

$$m \ \le\ \alpha \ln \sum_{i=1}^{n} e^{q_i/\alpha} \ \le\ m + \alpha \ln n \qquad \Longrightarrow \qquad \lim_{\alpha\downarrow 0}\ \alpha \ln \sum_{i=1}^{n} e^{q_i/\alpha} = \max_i q_i$$

by the squeeze rule; for $q = (1,\ 0.5,\ 0.9)$ the soft maximum is approximately 1.921, 1.032 and 1.0000005 at $\alpha = 1,\ 0.1,\ 0.01$. This is the temperature limit of entropy-regularised Bellman equations (Module 7, Primer E).

Convergence rates

DEFINITION — LINEAR, SUBLINEAR AND QUADRATIC CONVERGENCE

Errors $e_k \to 0$ converge **linearly** (geometrically) with rate $q \in (0,1)$ if $e_{k+1} \le q\, e_k$, so $e_k \le q^k e_0$; **sublinearly** if slower than every geometric rate, typically $e_k \le C/k^p$; **quadratically** if $e_{k+1} \le C e_k^2$ (Newton's method near a solution).

Reaching $e_k \le \varepsilon$ takes $k \ge \ln(e_0/\varepsilon)/\ln(1/q)$ iterations at a linear rate, growing like $\ln(1/\varepsilon)$, but $k \ge 1/\varepsilon^2$ for $1/\sqrt{k}$. For $\varepsilon = 10^{-6}$, $e_0 = 1$: 20 iterations at $q = 0.5$ against 10^{12}. On a logarithmic plot a linear rate is a straight line, a sublinear one flattens (see the explorer). Used for geometric Lyapunov decay and interior-point iteration counts $O(\sqrt{n}\log(1/\varepsilon))$ in Module 2, $O(1/\sqrt{T})$ primal-dual rates in Module 8, and gradient descent in Primer B.

With limits and the geometric series, Banach's existence proof (Section 2) takes six steps.

DERIVATION: BANACH'S FIXED-POINT THEOREM, STEP BY STEP

1. Setup | 2. Steps shrink | 3. Cauchy | 4. Limit | 5. Bounds | 6. Use it

Step 1: What is to be proved

$X \subseteq \mathbb{R}^n$ is nonempty and closed, $g : X \to X$, $\|g(x) - g(y)\| \le L\|x - y\|$ with $L \in [0,1)$, and $x_{k+1} = g(x_k)$ from any $x_0 \in X$. Example: $g(x) = 0.5x + 1$, $x_0 = 0$, so $x_1 = 1$, $x_2 = 1.5$, $x_3 = 1.75$.

$$\text{Goal:} \qquad x_k \to x^*, \qquad g(x^*) = x^*, \qquad \|x_k - x^*\| \le \frac{L^k}{1-L}\,\|x_1 - x_0\|.$$

KEY INSIGHT

We do not know x^* yet, so we control the distances between consecutive iterates, which we can compute.

← Previous Next →

Practice: Sequences, limits and series

Work from Easy to Medium before trying Hard. Use the review link for a missing definition, then try again with the explanation closed.

▼ **Exercise 0.4a — Easy: Turn a tolerance into an index**

Let $a_t = 3/(t+1)$ for integers $t \geq 0$. Write the first four terms, state the limit, and find the smallest T such that $|a_t| < 0.1$ for every $t \geq T$.

Review: Sequences, limits and series

▼ **Show hint**

Solve $3/(t+1) < 0.1$ first, then remember that t is an integer and the inequality is strict.

▼ **Show worked solution**

Compute: substituting $t = 0, 1, 2, 3$ gives $3,\ 3/2,\ 1,\ 3/4$. The numerator stays fixed while the denominator grows, so $a_t \to 0$.

Solve: the denominator is positive, so multiplying gives $3 < 0.1(t+1)$, hence $t+1 > 30$ and $t > 29$. Therefore the smallest integer threshold is $T = 30$.

Check the quantifier: for every $t \geq 30$, $3/(t+1) \leq 3/31 \approx 0.09677 < 0.1$. The preceding term is $a_{29} = 0.1$, so $T = 29$ fails the strict inequality.

▼ **Exercise 0.4b — Medium: Bound the part of a return you have not summed**

With discount $\gamma = 1/2$ and rewards $|r_t| \leq 2$, bound the absolute infinite return. Compute the first four terms and the omitted tail when $r_t = 2$ always. What is the smallest number T of terms guaranteeing a tail at most 0.01 for every permitted reward sequence?

Review: Sequences, limits and series

▼ **Show hint**

The first T terms have indices $0, \ldots, T-1$; the tail bound is $2\gamma^T/(1-\gamma)$.

▼ **Show worked solution**

Infinite bound: the triangle inequality gives $|\sum_{t\geq 0} 2^{-t} r_t| \leq 2\sum_{t\geq 0} 2^{-t} = 2/(1-1/2) = 4$. Constant rewards 2 attain this bound.

Four terms: $2 + 1 + 1/2 + 1/4 = 15/4 = 3.75$. The exact tail is $4 - 3.75 = 1/4$; the formula also gives $4 \cdot 2^{-4} = 1/4$.

Accuracy: require $4 \cdot 2^{-T} \leq 0.01$, or $2^T \geq 400$. Since $2^8 = 256 < 400 \leq 512 = 2^9$, $T = 9$. At $T = 8$ the worst tail is 0.015625; at $T = 9$ it is 0.0078125. This is an absolute error target, not a percentage of the return.

▼ **Exercise 0.4c — Hard: Extract convergence from a telescoping inequality**

Suppose $V_t \geq 0$, $V_0 = 3$, and $V_{t+1} - V_t \leq -\frac{1}{2}\|x_t\|^2$ for every $t \geq 0$. Prove $x_t \to 0$. Bound the total number of indices with $\|x_t\| \geq 0.2$. Does this give a time after which all such indices have disappeared?

Review: Sequences, limits and series

▼ **Show hint**

Sum through $t = T - 1$. Every index above the threshold contributes at least 0.2^2 to the sum of squared norms.

▼ **Show worked solution**

Telescope: adding the inequalities cancels all intermediate V_t and gives $\frac{1}{2}\sum_{t=0}^{T-1}\|x_t\|^2 \leq V_0 - V_T \leq 3$. Thus every partial sum is at most 6.

Conclude convergence: these partial sums are nondecreasing and bounded, so the nonnegative series $\sum_t \|x_t\|^2$ converges. Its terms must tend to 0; taking square roots gives $\|x_t\| \to 0$, which means $x_t \to 0$.

Count excursions: each index with $\|x_t\| \geq 0.2$ costs at least 0.04 of the budget 6, so there are at most $6/0.04 = 150$ such indices over all time.

Interpret the limit: only finitely many excursions means there is a last one, but the budget does not tell us its index. A single excursion could occur arbitrarily late while spending only 0.04. A numerical time bound needs more information about the dynamics.

5. Supremum, Infimum & Bounds

Worst cases over infinite sets (largest gradient, worst disturbance, best policy) are suprema; unlike maxima they always exist, possibly as $\pm\infty$.

DEFINITION — UPPER BOUND, SUPREMUM, MAXIMUM

For a nonempty $A \subseteq \mathbb{R}$, u is an **upper bound** if $a \leq u$ for all $a \in A$, and A is **bounded above** if it has one. When A is bounded above, its **supremum** $\sup A$ is the least upper bound: (i) $a \leq \sup A$ for all $a \in A$, and (ii) for every $\varepsilon > 0$ some $a \in A$ has $a > \sup A - \varepsilon$ (if A is not bounded above, $\sup A = +\infty$, see the next box). If $\sup A \in A$ it is the **maximum**. Infimum (greatest lower bound) and minimum are symmetric, $\inf A = -\sup(-A)$, and $\sup_{x \in X} f(x) := \sup f(X)$. *Completeness of* $\mathbb{R}$: every nonempty set that is bounded above has a supremum in $\mathbb{R}$.

Intuition and examples. A supremum is a ceiling you can approach but may never touch: $[0, 1)$ has supremum 1 and no maximum. $\sup_{r \geq 0}(1 - e^{-r}) = 1$ is not attained (a bounded continuity modulus in Module 5), while $\sup_x e^{-x^2} = 1$ is a maximum. The Lipschitz constant $L^\star(f) = \sup_{x \neq y} |f(x) - f(y)|/|x - y|$ of $\sin$ is 1, approached as $x, y \to 0$ but never attained. **To prove** $\sup A = s$, show that s is an upper bound and find $a_n \in A$ with $a_n \to s$.

> **DEFINITION — EXTENDED REALS AND OPTIMAL VALUES**
>
> $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$; $\sup A = +\infty$ if A is unbounded above, and by convention $\sup \emptyset = -\infty$, $\inf \emptyset = +\infty$ (every number bounds $\emptyset$). $a + \infty = \infty$ for $a \neq -\infty$, while $\infty - \infty$ is undefined. The value $\inf\{f(x) : x \in S\}$ of a minimization is $+\infty$ if it is infeasible ($S = \emptyset$) and $-\infty$ if it is unbounded below. A function with values in $\overline{\mathbb{R}}$ is **proper** if it never equals $-\infty$ and is finite somewhere.

Example. SafeOpt starts from $C_0(x) = [h, \infty)$ on the seed, so $l_0(x) = h$ and $u_0(x) = +\infty$, and from $C_0(x) = \mathbb{R}$ elsewhere ($l_0 = -\infty$). If data contradict an earlier band, $C_t(x) = \emptyset$, and the conventions would give $l_t = +\infty > u_t = -\infty$; Module 4 treats this as a failed confidence event or numerical error, not as a certificate; one empty intersection alone does not prove that the assumptions are false.

> **FACT — RULES FOR SUPREMA AND INFIMA**
>
> For nonempty bounded $A, B \subseteq \mathbb{R}$, $c > 0$ and functions f, g, ϕ: (1) $\sup(cA) = c \sup A$ and $\sup(-A) = -\inf A$; (2) $\sup(A + B) = \sup A + \sup B$, where $A + B = \{a + b\}$; (3) $\sup_x(f + g) \leq \sup_x f + \sup_x g$ and $\inf_x(f + g) \geq \inf_x f + \inf_x g$, possibly strictly; (4) $A \subseteq B \Rightarrow \sup A \leq \sup B$ and $\inf A \geq \inf B$: optimizing over a larger set (relaxing constraints) raises a maximum and lowers a minimum; (5) the **max-min inequality** $\sup_y \inf_x \phi(x, y) \leq \inf_x \sup_y \phi(x, y)$.
>
> *Proofs.* (2) "$\leq$": $a + b \leq \sup A + \sup B$. "$\geq$": pick $a > \sup A - \frac{\varepsilon}{2}$, $b > \sup B - \frac{\varepsilon}{2}$, so $\sup(A + B) > \sup A + \sup B - \varepsilon$ for every $\varepsilon > 0$. (1) is similar. (3): $f(x) + g(x) \leq \sup f + \sup g$ for every x; it is strict for $f(x) = x$, $g(x) = -x$ on $[0, 1]$ ($0 < 1$). (5): $\inf_x \phi(x, y) \leq \phi(x', y) \leq \sup_{y'} \phi(x', y')$ for all x', y; take $\sup_y$ on the left and $\inf_{x'}$ on the right. ■ For $\phi(x, y) = (x - y)^2$ on $\{0, 1\}^2$: $\inf_x \sup_y \phi = 1$ but $\sup_y \inf_x \phi = 0$.

WHERE THIS IS USED

$L^\star(f) = \sup_x \|J_f(x)\|$ in Module 12; "the infimum of a sum is at least the sum of the infima", optimal values $\pm\infty$ and weak duality via rule (5) in Module 2 and Module 8; $\ell_t = \inf C_t$, $u_t = \sup C_t$ in Module 5; argmax sets and min-max problems in Primer B.

PITFALL — WRITING MAX WHERE ONLY SUP IS KNOWN

"$\max_{x\in D} f(x)$" presumes a maximizer: fine for nonempty finite D, or nonempty compact D and continuous f (Section 6). On $(0,1)$, $\max x$ does not exist although $\sup x = 1$. A step "pick $x_t \in \arg\max$" needs that existence, or an explicit approximate choice.

Limit superior and limit inferior

$\limsup_t a_t := \lim_{T\to\infty} \sup_{t\geq T} a_t$ and $\liminf_t a_t := \lim_{T\to\infty} \inf_{t\geq T} a_t$ always exist in $\overline{\mathbb{R}}$, because $\sup_{t\geq T} a_t$ is nonincreasing and $\inf_{t\geq T} a_t$ nondecreasing in T; a sequence converges iff they are equal and finite. **Example.** Let $r_t = 1$ if $4^k \leq t < 2\cdot 4^k$ for some integer $k \geq 0$, else $r_t = 0$ ($t \geq 1$). The average $\frac{1}{T}\sum_{t=1}^{T} r_t$ approaches $\frac{2}{3}$ at the end of each block of ones and $\frac{1}{3}$ at the end of each block of zeros: $\liminf = \frac{1}{3}$, $\limsup = \frac{2}{3}$, no limit. A long-run constraint $\liminf_T \frac{1}{T}\sum_{t<T} r_t \geq c$ says that for every $\varepsilon > 0$ the average is at least $c - \varepsilon$ for all large T (it may stay slightly below c forever, like $c - \frac{1}{T}$), whether or not the average settles (Module 8).

Preview: almost everywhere and almost surely

DEFINITION — MEASURE ZERO, A.E., A.S., ESSENTIAL SUPREMUM

$N \subseteq \mathbb{R}^n$ has **measure zero** (zero volume) if for every $\varepsilon > 0$ it can be covered by countably many boxes of total volume below ε, e.g. finite sets in positive dimension, lines in $\mathbb{R}^2$, hyperplanes $\{x : w^\top x + b = 0\}$ with $w \neq 0$, and finite unions of these. A property holds **almost everywhere** (a.e.) if it fails only on a measure-zero set, and an event holds **almost surely** (a.s.) if its probability is 1. The **essential supremum** is the smallest almost-sure upper bound, $\operatorname{ess\,sup} Z = \inf\{c : \mathbb{P}(Z \leq c) = 1\}$.

Intuition and example. A ReLU network is differentiable except at the kinks where a hidden unit switches: finitely many pieces of hyperplanes $w^\top x + b = 0$ (whole hyperplanes for one hidden layer), a measure-zero set. Every Lipschitz function on $\mathbb{R}^n$ is differentiable a.e. (Rademacher). If U is uniform on $[0,1]$ and $Z = U$ except $Z = 10$ when $U = \frac{1}{2}$, then $\operatorname{ess\,sup} Z = 1$, although the supremum over all outcomes is 10. Almost-sure safety tolerates failures on a probability-zero set of disturbances; robust safety tolerates none (Primer C). Used in Module 1 (a.s. safety, $\operatorname{ess\,sup}$),

Module 12 and Module 10 (differentiable a.e.), Module 13 (gradient norm 1 almost everywhere) and Module 8 (a.s. convergence).

> **PITFALL — "PROBABILITY ZERO" IS NOT "IMPOSSIBLE"**
>
> A statement for almost every x says nothing about a particular x, and the exceptions may sit exactly where it matters (a boundary, a ReLU kink). Upgrading needs structure such as continuity: a continuous function that is ≥ 0 almost everywhere is ≥ 0 everywhere.

Practice: Suprema, infima and bounds

Work from Easy to Medium before trying Hard. Use the review link for a missing definition, then try again with the explanation closed.

▼ Exercise 0.5a — Easy: A bound need not belong to the set

For $A = (-2, 3]$, find $\inf A$, $\sup A$, the minimum and the maximum. Repeat for $B = \{1 - 1/n : n = 1, 2, \dots\}$. Explain any missing extrema.

Review: Suprema, infima and bounds

▼ Show hint

First identify the floor and ceiling; then ask whether each is actually an element.

▼ Show worked solution

Interval: every member of A is above -2 and at most 3. Values approach -2 from above, and 3 belongs to A. Hence $\inf A = -2$, $\sup A = 3$, no minimum, and maximum 3.

Sequence of values: B begins $0, 1/2, 2/3, 3/4, \dots$. Every member is nonnegative, and 0 is included at $n = 1$, so the infimum and minimum are both 0.

Upper bound: $1 - 1/n < 1$ for every finite n, but these values tend to 1. Therefore $\sup B = 1$ and no maximum exists. To check leastness explicitly, for any $\varepsilon > 0$ choose $n > 1/\varepsilon$; then $1 - 1/n > 1 - \varepsilon$.

▼ **Exercise 0.5b — Medium: Separate a supremum from a limit superior**

For $a_t = (-1)^t + 1/(t+1)$, $t \geq 0$, compute $\sup_t a_t$, $\inf_t a_t$, $\limsup_t a_t$ and $\liminf_t a_t$. Which bounds are attained, and does the sequence converge?

Review: Suprema, infima and bounds

▼ **Show hint**

Split into even and odd indices. A limit superior ignores any fixed finite beginning.

▼ **Show worked solution**

> **Even indices:** $a_{2k} = 1 + 1/(2k+1)$ decreases from 2 towards 1. Hence the supremum of the entire sequence is 2, attained at $t = 0$.
>
> **Odd indices:** $a_{2k+1} = -1 + 1/(2k+2)$ is always above -1 and decreases towards it. The infimum is -1, never attained.
>
> **Tails:** every tail still contains infinitely many even and odd indices. Their limiting upper and lower values are 1 and -1, respectively, so $\limsup a_t = 1$ and $\liminf a_t = -1$.
>
> **Convergence check:** since the two tail limits differ, the sequence has no single limit. The isolated initial value 2 affects the supremum but not the limit superior.

▼ **Exercise 0.5c — Hard: Compute both orders of a minimax problem**

Let $x, y \in \{0, 1\}$ and let $\phi(0,0) = 0, \phi(0,1) = 2, \phi(1,0) = 1, \phi(1,1) = 0$. Compute $\min_x \max_y \phi(x,y)$ and $\max_y \min_x \phi(x,y)$. Explain why changing the order changes the result.

Review: Suprema, infima and bounds

▼ **Show hint**

Treat x as a row and y as a column. For min-max take row maxima first; for max-min take column minima first.

▼ **Show worked solution**

Commit x first: row $x = 0$ has worst value $\max(0, 2) = 2$; row $x = 1$ has worst value $\max(1, 0) = 1$. The minimising player chooses $x = 1$, giving $\min_x \max_y \phi = 1$.

Commit y first: column $y = 0$ has best response $\min(0, 1) = 0$; column $y = 1$ has best response $\min(2, 0) = 0$. Thus $\max_y \min_x \phi = 0$.

Reason: in each expression the second player can respond to the first player's choice. The minimiser gains that advantage in max-min, producing $0 \leq 1$, as the general max-min inequality predicts. Equality requires additional structure; a finite set by itself does not supply it.

6. Open, Closed & Compact Sets; Continuity

Safety is decided at the boundary of the safe set; optimizers and positive margins exist thanks to compactness. We work in $\mathbb{R}^n$ with the Euclidean distance (other norms give the same open, closed and compact sets, Primer A).

DEFINITION — BALLS, INTERIOR, OPEN, CLOSED, CLOSURE, BOUNDARY, NEIGHBOURHOOD

$B_r(x) = \{y : \|y - x\| < r\}$ is the open ball, $\bar{B}_r(x) = \{y : \|y - x\| \leq r\}$ the closed ball. x is an **interior point** of A if $B_r(x) \subseteq A$ for some $r > 0$; $\operatorname{Int} A$ collects them. A is **open** if $A = \operatorname{Int} A$, and **closed** if its complement is open, equivalently if every convergent sequence in A has its limit in A. The **closure** $\operatorname{cl} A$ adds all such limits, and the **boundary** $\partial A = \operatorname{cl} A \setminus \operatorname{Int} A$ consists of the points every ball around which meets both A and its complement. A **neighbourhood** of x contains a ball around x; "locally" means "on some neighbourhood". Example: $[0, 1)$ has interior $(0, 1)$, closure $[0, 1]$, boundary $\{0, 1\}$, and is neither open nor closed.

FACT — SETS DEFINED BY A CONTINUOUS FUNCTION

If $h : \mathbb{R}^n \to \mathbb{R}$ is continuous and $C = \{h \geq 0\}$, then C and $\{h = 0\}$ are closed, $\{h > 0\}$ is open, $\{h > 0\} \subseteq \operatorname{Int} C$ and $\partial C \subseteq \{h = 0\}$. *Proof.* If $x_t \to x$ with $h(x_t) \geq 0$, then $h(x) = \lim h(x_t) \geq 0$; likewise for $\{h \leq 0\}$ and $\{h = 0\}$. The complement $\{h > 0\}$ of $\{h \leq 0\}$ is open and inside C, hence inside $\operatorname{Int} C$, and $\partial C = C \setminus \operatorname{Int} C \subseteq \{h = 0\}$. ■

When is the boundary the zero level set? If h is continuously differentiable (the partial derivatives in $\nabla h = (\partial h/\partial x_1, \ldots, \partial h/\partial x_n)$ exist and are continuous) and $\nabla h(x) \neq 0$ wherever $h(x) = 0$ (0 *is a regular value*), then $\partial C = \{h = 0\}$ and $\operatorname{Int} C = \{h > 0\}$. Reason: take x_0 with $h(x_0) = 0$ and follow the gradient, $\varphi(s) = h(x_0 + s\nabla h(x_0))$. By the chain rule (Primer B), $\varphi(0) = 0$ and $\varphi'(0) = \|\nabla h(x_0)\|^2 > 0$, so $\varphi < 0$ for small $s < 0$: every ball around x_0 contains points with $h < 0$, hence $x_0 \notin \operatorname{Int} C$, and $x_0 \in \partial C$ because C is closed. Without regularity this fails: $h(x) = x^2$ gives $C = \mathbb{R}$ and $\partial C = \emptyset$ although $h(0) = 0$; and $h(x) = -x^3$ with $\dot{x} = 1$ passes the boundary check $\dot{h} = -3x^2\dot{x} = 0 \geq 0$ at 0, yet the state leaves $C = (-\infty, 0]$ at once. **Why the boundary matters:** a continuous trajectory that starts in C and ends outside must cross ∂C (intermediate value theorem for $t \mapsto h(x(t))$), so, for $\dot{x} = f(x)$ with $f : \mathbb{R}^n \to \mathbb{R}^n$ locally Lipschitz, forward invariance on each maximal solution interval can be checked on a regular boundary alone (Nagumo, Module 10).

Going deeper — tangent cones and the Hausdorff distance ▾

With $\operatorname{dist}(y, C) = \inf_{c \in C} \|y - c\|$, the **tangent cone** of a closed C at x is $T_C(x) = \{v : \liminf_{\tau \downarrow 0} \operatorname{dist}(x + \tau v, C)/\tau = 0\}$, which equals $\{v : \nabla h(x)^\top v \geq 0\}$ at a regular boundary point; Nagumo: C is forward invariant iff $f(x) \in T_C(x)$ on ∂C (unique solutions). The **Hausdorff distance** $d_H(A, B) = \max\{\sup_{a \in A} \operatorname{dist}(a, B), \sup_{b \in B} \operatorname{dist}(b, A)\}$ of nonempty compact sets ($d_H([0, 1], [0, 1.5]) = 0.5$) measures the convergence of reachable sets in Module 10.

Compact sets and the extreme value theorem

DEFINITION AND THEOREM — COMPACTNESS (HEINE–BOREL, BOLZANO–WEIERSTRASS)

$A \subseteq \mathbb{R}^n$ is **bounded** if $A \subseteq B_R(0)$ for some R, and **compact** if every sequence in A has a subsequence ($a_{t_1}, a_{t_2}, \ldots$ with $t_1 < t_2 < \cdots$) converging to a point of A. In $\mathbb{R}^n$: compact $\iff$ closed and bounded, every bounded sequence has a convergent subsequence, and a compact set is covered by finitely many balls of any radius $\tau > 0$. *Proof idea:* halve a box containing the sequence repeatedly, keeping a half with infinitely many terms; the boxes shrink to the limit of a subsequence, and closedness keeps it in the set.

Examples. $[0,1]^d$, closed balls and ellipsoids $\{x : x^\top P x \le c\}$ with $P \succ 0$ (bounded since $x^\top P x \ge \lambda_{\min}(P)\|x\|^2$, Primer A) are compact; $\mathbb{R}^n$ for $n \ge 1$, $(0,1]$ and $\{x : x_1^2 - x_2^2 \le 1\}$ (closed, unbounded) are not. Sublevel sets need care: $V(x) = x^2/(1+x^2)$ is positive away from 0, and $\{V \le 0.5\} = [-1,1]$, but $\{V \le c\} = \mathbb{R}$ for every $c \ge 1$.

THEOREM — WEIERSTRASS EXTREME VALUE THEOREM

A continuous $f : K \to \mathbb{R}$ on a nonempty compact $K \subseteq \mathbb{R}^n$ attains its maximum and minimum. *Proof.* Let $M = \sup_K f \in \overline{\mathbb{R}}$ and $x_t \in K$ with $f(x_t) \to M$. A subsequence converges to some $\bar{x} \in K$, and continuity gives $f(\bar{x}) = M$, so M is finite and attained; use $-f$ for the minimum. ■ *Why each assumption matters:* x on $(0,1]$ never attains $\inf = 0$ (not closed); e^{-x} on $[0,\infty)$ neither (unbounded); $f(x) = x$ on $[0,1)$ with $f(1) = 0$ never attains $\sup = 1$ (discontinuous).

Coercivity. If S is closed and nonempty, f continuous and $f(x) \to \infty$ as $\|x\| \to \infty$, then f has a minimizer on S (Weierstrass on the compact set $\{x \in S : f(x) \le f(x_0)\}$). So the safety filter $\min_u \|u - u_{\text{nom}}\|^2$ subject to $a + b^\top u \ge 0$ has a solution on an unbounded feasible set, whereas $\max_{u \in \mathbb{R}^m} b^\top u$ ($b \ne 0$) has none. **Positive margins:** if V is continuous, $V(0) = 0$ and $V > 0$ elsewhere, then $\min_K V > 0$ on every nonempty compact K with $0 \notin K$.

Proof — strict decrease on a compact sublevel set forces convergence (the argument of Module 11) ▾

Claim. Let F be continuous with $F(0) = 0$, V continuous with $V(0) = 0$ and $V(x) > 0$ for $x \ne 0$, $\mathcal{V}(c) = \{V \le c\}$ compact, and $V(F(x)) < V(x)$ on $\mathcal{V}(c) \setminus \{0\}$. Then trajectories of $x_{t+1} = F(x_t)$ from $x_0 \in \mathcal{V}(c)$ stay in $\mathcal{V}(c)$ and converge to 0.

(1) Induction: $x_t \in \mathcal{V}(c)$ gives $V(x_{t+1}) \le V(x_t) \le c$ ($F(0) = 0$ covers $x_t = 0$). *(2) Monotone convergence:* $v_t = V(x_t) \downarrow v^* \ge 0$. *(3) Contradiction and compactness:* if $v^* > 0$, all x_t lie in the

compact $K = \{v^* \le V \le c\}$, which excludes 0; the continuous $\Delta V(x) = V(F(x)) - V(x)$ is negative there, so $\max_K \Delta V = -\eta < 0$ and $v_t \le v_0 - t\eta \to -\infty$, impossible. *(4)* For $\varepsilon > 0$, $m_\varepsilon = \min\{V(x) : x \in \mathcal{V}(c), \|x\| \ge \varepsilon\} > 0$ (if that compact set is nonempty), and $v_t < m_\varepsilon$ forces $\|x_t\| < \varepsilon$. ■

Audit. Induction, monotone convergence, suprema, Weierstrass (twice) and contradiction all appear. Without continuity of F step (3) fails (Module 11: $x_t = 1 + 2^{-t} \to 1$). The argument gives no rate: for $F(x) = x/(1+x^2)$, $V = x^2$, one finds $\eta = v^*\left(1 - (1+v^*)^{-2}\right) \to 0$ as $v^* \to 0$, matching the sublinear $1/\sqrt{2t}$ of Section 2.

Uniform continuity, nets, and gridding with a Lipschitz margin

f is **uniformly continuous** on X if $\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in X : \|x - y\| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$: one δ for all points (for continuity δ may depend on the point). Continuous functions on compact sets are uniformly continuous (Heine–Cantor), and L-Lipschitz ones are too, with $\delta = \varepsilon/L$ for $L > 0$ (any δ works if $L = 0$: f is constant); $1/x$ on $(0, 1]$ and x^2 on $\mathbb{R}$ are continuous but not uniformly so.

DEFINITION AND FACT — T-NETS, COVERING NUMBERS, A LIPSCHITZ MARGIN

For $\tau > 0$, a finite $Z \subseteq X$ is a **τ-net** if every $x \in X$ has some $z \in Z$ with $\|x - z\| \le \tau$; the **covering number** $M(\tau, X)$ is the smallest size of one. For $d \ge 1$, cubes of side $2\tau/\sqrt{d}$ have half-diagonal τ, so $M(\tau, [0,1]^d) \le \lceil \sqrt{d}/(2\tau) \rceil^d$: for $\tau = 0.1$, 64 points in $d = 2$ but $16^{10} \approx 1.1 \cdot 10^{12}$ in $d = 10$. **Fact:** if h is L-Lipschitz, Z is a τ-net of X and $h(z) \ge L\tau$ on Z, then $h \ge 0$ on X, because $h(x) \ge h(z) - L\|x - z\| \ge L\tau - L\tau = 0$. ■

Example. $h(x) = 1 - x^2$ on $X = [-0.95, 0.95]$ has $L = \max_X |h'| = 1.9$. The 20 points $-0.95, -0.85, \ldots, 0.95$ form a 0.05-net; the margin $L\tau = 0.095$ is cleared by the smallest grid value $h(\pm 0.95) = 0.0975$, so $h \ge 0$ on X is certified. Spacing 0.2 ($\tau = 0.1$, margin 0.19) fails although $h \ge 0$ is true: finer grids certify more, at a cost growing like τ^{-d} (Section 7).

WHERE THIS IS USED

Open domains, $\operatorname{Int} C$, ∂C in Module 10; regular values in Module 1 and Module 7; closed failure sets in Module 7; compact sublevel sets in Module 11 and Module 14; compact domains in Module 3; attained optima in Module 7 and Module 10, and attainment failing in Module 8; covering numbers in Module 3; ϵ-nets in Module 10; grids in Module 5.

Connectedness, semicontinuity and density

Connectedness. A is *path-connected* if any two points are joined by a continuous path inside A; *path components* are the maximal path-connected subsets (for open sets they coincide with the usual connected components, in general not). Continuous images of path-connected sets are path-connected, so a continuous parameterisation from $\mathbb{R}^N$ reaches only one path component of a set that has several. A finite grid in Euclidean space has no continuous path joining distinct points; its connectivity depends on an adjacency rule, and certified reachability is a third notion (Module 6, Module 4, Module 13, Module 11).

Semicontinuity. f is *lower semicontinuous* if $f(x) \le \liminf_t f(x_t)$ whenever $x_t \to x$, equivalently if every $\{f \le c\}$ is closed; the Weierstrass argument still yields minima on nonempty compact sets. Example: $f = 0$ on $(-\infty, 0]$, $f = 1$ on $(0, \infty)$ (Module 8).

Density. A class $\mathcal{G}$ is dense in $\mathcal{F}$ in the sup norm on a compact X if $\forall f \in \mathcal{F}\ \forall \varepsilon > 0\ \exists g \in \mathcal{G}$: $\sup_X |f - g| \le \varepsilon$: arbitrarily good approximation, not exact representation (Module 13).

Practice: Topology, existence and grid certificates

Work from Easy to Medium before trying Hard. Use the review link for a missing definition, then try again with the explanation closed.

▼ **Exercise 0.6a — Easy: Describe an interval all the way to its boundary**

Working in $\mathbb{R}$, find the interior, closure and boundary of $A = (-1, 1]$. Is A open, closed, bounded or compact? Is $B = \{0, 1\}$ compact?

Review: Topology, existence and grid certificates

▼ **Show hint**

An interior point needs a small interval around it inside the set. A closed set keeps every limit of its convergent sequences.

▼ **Show worked solution**

> **Interior:** every x strictly between -1 and 1 has a small interval around it in A. At 1, every such interval extends above 1, so $\operatorname{Int} A = (-1, 1)$.
>
> **Closure and boundary:** adding the missing limit -1 gives $\operatorname{cl} A = [-1, 1]$. Removing the interior leaves $\partial A = \{-1, 1\}$. A boundary point need not belong to the original set.
>
> **Classify:** A is not open (it contains noninterior point 1), not closed (a sequence $-1 + 1/n$ in A tends to missing point -1), and bounded. In $\mathbb{R}$, compact means closed and bounded, so A is not compact. The finite set B is closed and bounded and therefore compact, even though it has no interior.

▼ Exercise 0.6b — Medium: Decide whether a smallest value is attained

For $f(x) = x^2$, decide whether a minimum exists on each of $(0, 1]$, $[0, 1]$ and $\mathbb{R}$. State which cases follow directly from the extreme value theorem and explain the remaining cases.

Review: Topology, existence and grid certificates

▼ Show hint

Find the infimum first. Compactness is sufficient for a continuous function to attain it, but not necessary.

▼ Show worked solution

On $(0, 1]$: $f(x) > 0$ for every permitted x, while $f(1/n) = 1/n^2 \to 0$. The infimum is 0 but there is no minimiser. The domain is bounded but not closed.

On $[0, 1]$: the continuous function is on a nonempty compact interval, so the theorem applies. Explicitly $f(0) = 0 \leq f(x)$, giving the unique minimiser 0.

On $\mathbb{R}$: the domain is not compact, so that theorem alone does not apply. Nevertheless $x^2 \geq 0$ with equality only at 0, so a unique minimum exists. Coercivity ($x^2 \to \infty$ as $|x| \to \infty$) is another route to existence here. Failure of a sufficient condition is not proof of nonexistence.

▼ **Exercise 0.6c — Hard: Certify the space between measurements**

An unknown 3-Lipschitz function $h : [0, 1] \to \mathbb{R}$ satisfies $h(g) \geq 0.35$ at $G = \{0, 0.2, 0.4, 0.6, 0.8, 1\}$. Prove a uniform lower bound. Would measurements at $\{0, 0.4, 0.8, 1\}$ with the same lower bound suffice? Give a counterexample if not.

Review: Topology, existence and grid certificates

▼ **Show hint**

Use half the largest gap as the covering radius. For the counterexample try $h(x) = 0.35 - 3\,\mathrm{dist}(x, G)$.

▼ **Show worked solution**

> **Fine grid:** the gaps are 0.2, so each point is within 0.1 of a grid point. For a nearest g, $h(x) \geq h(g) - 3|x - g| \geq 0.35 - 0.3 = 0.05$. Thus every point has a positive margin of at least 0.05.
>
> **Coarse grid:** the largest gap is 0.4, so the covering radius is 0.2. The same argument gives only $0.35 - 0.6 = -0.25$, which does not certify nonnegativity.
>
> **Show the limitation is real:** let $G_c = \{0, 0.4, 0.8, 1\}$ and $h_c(x) = 0.35 - 3\,\mathrm{dist}(x, G_c)$. Distance to a nonempty set is 1-Lipschitz: comparing to a nearest point and using the triangle inequality bounds its change by $|x - y|$. Therefore h_c is 3-Lipschitz and equals 0.35 on the grid. At $x = 0.2$, its value is $0.35 - 3(0.2) = -0.25$. Positive samples alone do not prove safety between them.

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7. Asymptotic Notation & Rates

Rates and costs are stated up to constants ("$\gamma_T = O((\log T)^{d+1})$" for the information gain of a GP, Module 3, where γ_T is not a discount factor; "an $O(t^3)$ determinant"); here is what that promises.

> **DEFINITION — O, Ω, Θ, o, $\tilde{O}$**
>
> For $f, g \geq 0$ as $n \to \infty$: $f = O(g)$ if $\exists C > 0, n_0$ with $f(n) \leq Cg(n)$ for all $n \geq n_0$; $f = \Omega(g)$ if $\exists c > 0, n_0$ with $f(n) \geq cg(n)$ for $n \geq n_0$; $f = \Theta(g)$ if both; $f = o(g)$ if $\forall \varepsilon > 0\ \exists n_0\ \forall n \geq n_0 :\ f(n) \leq \varepsilon g(n)$, equivalently $f(n)/g(n) \to 0$ when g is eventually positive; $f = \tilde{O}(g)$ if $f(n) = O\big(g(n)\,(\log n)^k\big)$ for some constant $k \geq 0$ (so $\log n = \tilde{O}(1)$; with several parameters, papers differ in which logarithms they hide). For a small parameter, "$f(x) = O(g(x))$ as $x \to 0$" means $|f(x)| \leq C|g(x)|$ for all small $|x|$, e.g. $e^x = 1 + x + O(x^2)$. Always ask which variable tends where, what is held fixed, and what the constant may depend on (dimension, δ, γ, a norm bound).

Examples. $3n^2 + 10n + 5 = \Theta(n^2)$: for $n \geq 10$, $10n \leq n^2$ and $5 \leq n^2$, so $3n^2 \leq 3n^2 + 10n + 5 \leq 5n^2$. $\ln n = o(n^a)$ and $n^k = o(e^{an})$ for all $a > 0$, k. Module 12 writes $\log\det(N + \varepsilon E) = \log\det N + \varepsilon\,\mathrm{tr}(N^{-1}E) + O(\varepsilon^2)$ as $\varepsilon \to 0$, and Module 2 an error "within $O(\eta)$", i.e. at most $C\eta$ for small step sizes η.

T	10^2	10^4	10^6	10^8	10^{10}
$\sqrt{T}$	10	100	1000	10^4	10^5
$T^{2/3}$	≈ 21.5	≈ 464	10^4	$\approx 2.15 \cdot 10^5$	$\approx 4.64 \cdot 10^6$
$(\ln T)^3$	≈ 97.7	≈ 781	≈ 2637	≈ 6251	≈ 12208

Comparing rates. $(\ln T)^3 = o(\sqrt{T})$ and $\sqrt{T} = o(T^{2/3})$, yet $(\ln T)^3$ exceeds $\sqrt{T}$ for every T from 4 to about $2.4 \cdot 10^7$: an o-statement says nothing about a particular T.

Operation counts. A dense Cholesky factorisation, determinant or solve of a $t \times t$ matrix costs $O(t^3)$ operations (about $t^3/3$ for Cholesky), a matrix-vector product $O(t^2)$; doubling t multiplies $O(t^3)$ by about 8. Total cost is iterations times cost per iteration: interior-point SDP solvers need $O(\sqrt{n}\log(1/\varepsilon))$ iterations of $O(mn^3 + m^2n^2 + m^3)$ each. O predicts growth, not seconds. Grids explode with dimension: m points per axis give m^d points, e.g. $100^3 = 10^6$ but $100^6 = 10^{12}$.

> **DEFINITION — POLYNOMIAL TIME, P, NP, NP-HARD (WORKING VERSION)**
>
> A **decision problem** asks a yes/no question about an input of n bits; it is in **P** if one algorithm answers every instance in $O(n^k)$ steps for a fixed k. **NP** contains the problems whose "yes" answers have certificates checkable in polynomial time (a claimed counterexample to robustness is checked by one forward pass). A problem is **NP-hard** if every NP problem reduces to it in polynomial time, and **NP-complete** if it is also in NP. Whether P $=$ NP is open; if P $\neq$ NP, no polynomial-time algorithm solves all instances of an NP-hard problem. It is a worst-case statement about growing sizes, not a claim that every instance is hard; it is why the modules use relaxations (LipSDP) and branch-and-bound verifiers.

DEFINITION — SAMPLE COMPLEXITY, CUMULATIVE AND SIMPLE REGRET

The **sample complexity** $n(\varepsilon, \delta)$ is the number of samples that guarantees accuracy ε with probability at least $1 - \delta$. For queries $x_1, \dots, x_T$ and a comparator x^* (e.g. a maximizer of f), the **cumulative regret** is $R_T = \sum_{t=1}^{T} \big(f(x^*) - f(x_t)\big)$, and the **simple regret** of a recommendation $\hat{x}_T$ is $f(x^*) - f(\hat{x}_T)$. Regret is **sublinear** if $R_T = o(T)$, i.e. $R_T/T \to 0$ ("no regret"); then $\min_{t \le T} \big(f(x^*) - f(x_t)\big) \le R_T/T$.

Examples. By Hoeffding's inequality (Primer C), the mean of n independent $[0,1]$-valued samples is within ε of its expectation with probability $1 - \delta$ once $n \ge \ln(2/\delta)/(2\varepsilon^2)$, a sample complexity $O(\varepsilon^{-2} \log(1/\delta))$: 738 samples for $\varepsilon = \delta = 0.05$. $R_T \le 4\sqrt{T\gamma_T}$ with $\gamma_T = (\ln T)^2$ gives $R_T/T \le 4 \ln T/\sqrt{T} \to 0$ (sublinear), yet this bound is 0.37 at $T = 10^4$ and reaches 0.1 only at $T = 246{,}639$. Check products as a whole: Module 6 needs $\beta_N^2 \gamma_N = o(N)$, and $\sqrt{N} \cdot \sqrt{N} = N$ although each factor is $o(N)$. Bandits and regret are developed in Primer E.

WHERE THIS IS USED

Information-gain rates and $\tilde{O}$ regret in Module 3 and Module 3; $\gamma_t = O((\log t)^{d+1})$ and sample complexity in Module 4; $O(t^3)$ and hopeless grids in Module 5 and Module 5; $O(s)$ per backup in Module 6; $O(1/\sqrt{T})$ in Module 8; $O(\sqrt{\delta})$ in Module 9; solver costs in Module 2, Module 12 and Module 14; NP-hardness in Module 12 and Module 15; $O(n^2 c^3)$ in Module 13; $O(m^2 n^3)$ in Module 15.

PITFALL — A RATE IS NOT A CERTIFICATE

"$R_T = O(\sqrt{T})$" hides a constant and a threshold n_0; a guarantee at a specific T needs explicit constants. And the symbol has two uses: $O(\varepsilon^2)$ as $\varepsilon \to 0$ (a small remainder) and $O(n^2)$ as $n \to \infty$ (a growing cost).

Practice: Rates, constants and computational budgets

Work from Easy to Medium before trying Hard. Use the review link for a missing definition, then try again with the explanation closed.

▼ Exercise 0.7a — Easy: Prove a growth class with actual constants

For $f(n) = 2n^2 + 3n + 1, n \geq 1$, prove $f(n) = \Theta(n^2)$ using explicit constants. Is $f(n) = o(n^2)$?

Review: Rates, constants and computational budgets

▼ Show hint

For $n \geq 1$, both n and 1 are at most n^2. Little-o asks for a ratio tending to zero.

▼ Show worked solution

Lower bound: the extra terms are nonnegative, so $f(n) \geq 2n^2$. This proves $\Omega(n^2)$ with constant $c = 2$ and threshold $n_0 = 1$.

Upper bound: $3n \leq 3n^2$ and $1 \leq n^2$, hence $f(n) \leq (2 + 3 + 1)n^2 = 6n^2$. This proves $O(n^2)$ with $C = 6$ at the same threshold. Having both bounds gives $\Theta(n^2)$.

Ratio: $f(n)/n^2 = 2 + 3/n + 1/n^2 \to 2$, not 0. Thus f is not $o(n^2)$. Big-O permits a nonzero limiting ratio; little-o does not.

▼ Exercise 0.7b — Medium: Translate accuracy into work

Algorithm A guarantees error at most $4/\sqrt{T}$ after $T \geq 1$ iterations, each costing 100 operations. Algorithm B guarantees error at most $8/T$, each iteration costing 1000 operations. Find sufficient iteration and operation budgets for error at most 0.1. Which is cheaper under these bounds?

Review: Rates, constants and computational budgets

▼ Show hint

Solve each explicit inequality for T, round upward, then multiply by the cost per iteration.

▼ Show worked solution

A: $4/\sqrt{T} \leq 0.1$ implies $\sqrt{T} \geq 40$, hence $T \geq 1600$. The sufficient work budget is $1600 \cdot 100 = 160{,}000$ operations.

B: $8/T \leq 0.1$ implies $T \geq 80$. Its sufficient work budget is $80 \cdot 1000 = 80{,}000$ operations.

Compare carefully: B needs half as many operations under the stated guarantees, although each iteration costs ten times more. These are sufficient worst-case budgets; observed errors could be smaller. A rate without its constants and per-iteration cost would not settle this comparison.

▼ **Exercise 0.7c — Hard: Combine two rates without losing their meaning**

Suppose $0 \le R_T \le 5\sqrt{T}\log T$ for $T \ge 2$, where R_T is cumulative regret and $\log$ is natural. Prove the average regret tends to zero and give the resulting upper bound at $T = 10^4$. Separately, show why $a_T = o(T)$ and $b_T = o(T)$ do not imply $a_T b_T = o(T)$.

Review: Rates, constants and computational budgets

▼ **Show hint**

Divide cumulative regret by T. For the second claim use two square-root sequences.

▼ **Show worked solution**

Average: $0 \le R_T/T \le 5\log T/\sqrt{T}$. Since $\log T = o(T^{1/2})$, the right side tends to 0. The squeeze theorem therefore gives $R_T/T \to 0$, so the regret is sublinear; individual trial regrets need not all tend to zero.

Finite budget: at $T = 10^4$, $\sqrt{T} = 100$ and $\log T \approx 9.21034$, giving $R_T/T \le 0.46052$. An asymptotically vanishing bound can still be substantial at a particular budget.

Product counterexample: set $a_T = b_T = \sqrt{T}$. Each ratio $a_T/T = b_T/T = 1/\sqrt{T}$ tends to 0, but $a_T b_T = T$ and its ratio to T is 1. Products require multiplying the full expressions and then testing the resulting ratio.

Interactive: Fixed-Point Iteration & Geometric Series

Iterate $x_{k+1} = g(x_k)$ for several maps (cobweb and log-scale error against Banach's a priori bound) and sum the discounted series $\sum_t \gamma^t$.

FIXED-POINT ITERATION & GEOMETRIC SERIES

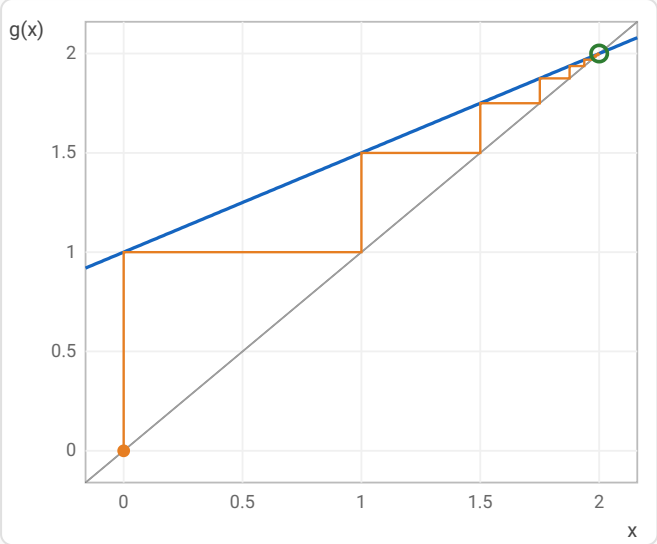
Map: affine: g(x) = a x + 1

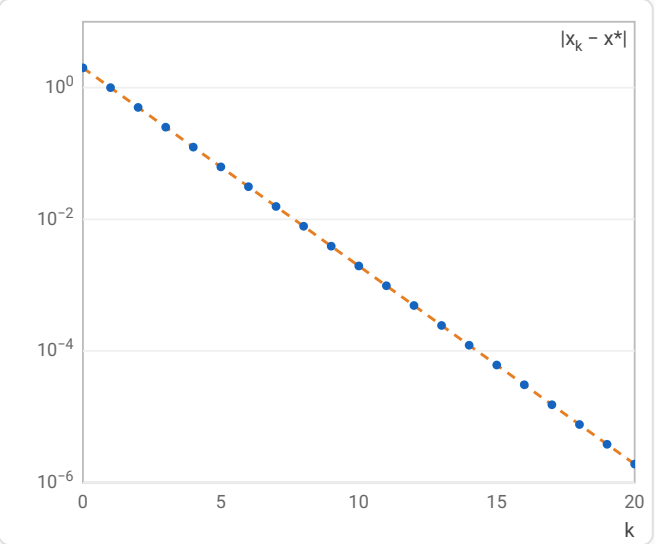
slope a 0.500 | start x_0 0.00 | iterations N 20

Fixed point x* = **2**; x_{20} = 2, error **$1.91\cdot10^{-6}$**. L = 0.5, a priori bound $1.91\cdot10^{-6}$.

Iterations to error ≤ 10^{-6}: **21** needed, **21** guaranteed by the bound.

A contraction on ℝ with L = |a|. The bound is exact; from x_0 = 0 the iterates are the sums 1 + a + … + a^{k-1}.



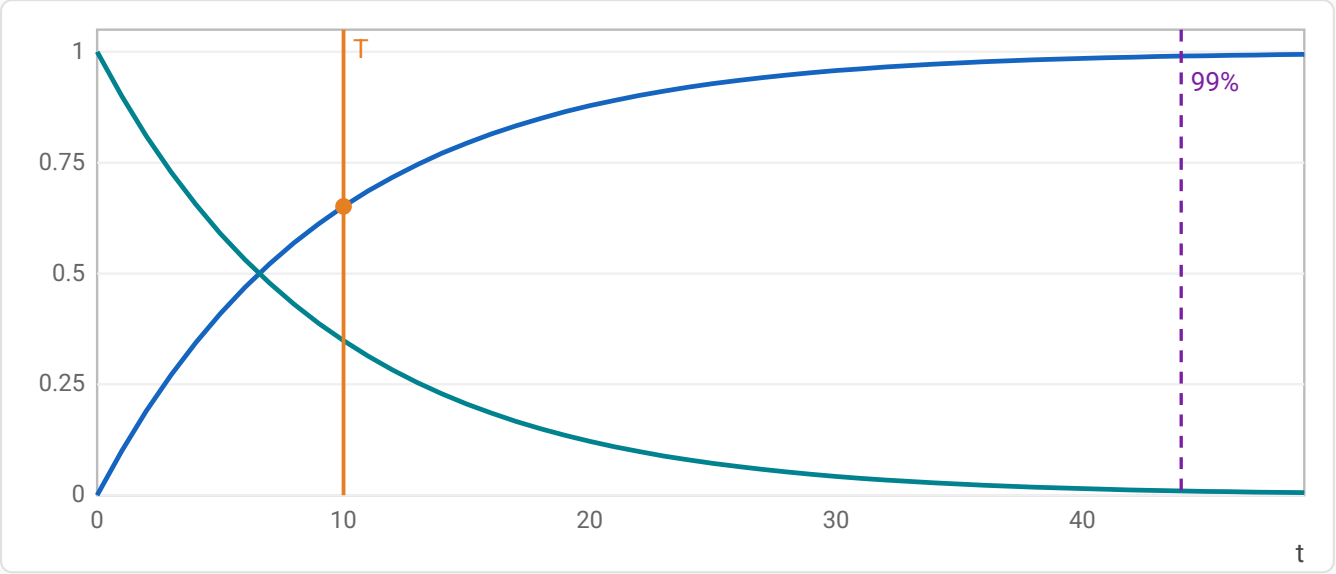


Cobweb: g (blue), $y = x$ (grey), staircase $(x_0, x_0) \to (x_0, x_1) \to (x_1, x_1) \to \cdots$ (orange), fixed point (green). Errors: $|x_k - x^*|$ (dots) and, if a factor $L < 1$ is known, the bound $L^{k-k_0}|x_{k_0+1} - x_{k_0}|/(1-L)$ (dashed).

discount γ **0.900** horizon T **10**

Run V ← 1 + γV above

Limit 1/(1 − γ) = **10**. $S_T = (1-\gamma^T)/(1-\gamma)$ = **6.5132** (65.1% of the limit), tail $\gamma^T/(1-\gamma)$ = 3.4868. 99% needs T ≥ **44**; the rule T ≥ ln(100)/(1 − γ) gives 47.



Blue: fraction $1-\gamma^t$ of the limit collected by t terms; teal: the remaining fraction γ^t; orange: horizon T; purple: first T with $\gamma^T \le 0.01$.

What to look for. (1) Affine, $a = 0.5$: the dots lie on the (exact) bound; $a = 0.99$ needs 1833 iterations; $a < 0$ spirals; $|a| \ge 1$ breaks the theorem ($a = 1$: no fixed point; $a = -1$: nonfixed starts alternate; $|a| > 1$: nonfixed starts run away; for $a \ne 1$, a start at $x^* = 1/(1-a)$ stays

fixed). (2) Cosine from $x_0 = 1$: the bound asks for 87 iterations, 32 suffice (slope 0.674 near x^* versus $L = 0.841$). (3) Logistic: $r = 2.8$ converges without a global contraction, $r = 2$ quadratically, $r = 3.2,\ 3.5,\ 3.56$ end on (approximate) cycles of period $2, 4, 8$, and at $r = 3.9$ the orbit is chaotic: no short cycle shows up. (4) Slow map: the dots bend, sublinear convergence. (5) The button's iterates V_k are the partial sums below.

From the mathematics to a real decision

A useful mathematical answer changes what someone can responsibly do. A temperature reading may support accepting a chamber, choosing a cooling command, or waiting for another solver iteration. Those are different decisions, and they require different statements. This chapter follows a constructed thermal-chamber model from a single reading to a claim about every future step. The numbers are teaching choices, not measurements from a particular machine. Their purpose is to make the order of the reasoning visible.

LEARNING OBJECTIVES

- Translate an acceptance rule into containment of sets and explain what a failed test means.
- Place control choices and unknown disturbances in the correct quantifier order.
- Build an all-time conclusion from a one-step bound without assuming the conclusion.
- Turn an error recurrence into a finite iteration budget, including its unavoidable error floor.

Begin with the decision and its units

Let x be temperature above a chosen reference, measured in degrees Celsius, and let the allowed interval be $C = [0, 5]$. This is a specification: for this hypothetical task a temperature deviation in that interval is acceptable. It is not a claim that the interval is physically appropriate for any actual chamber. A sensor reading y comes with a deterministic error bound $|x - y| \le \varepsilon$. The word deterministic means that every error in the interval is permitted; no probability distribution has been assumed.

The possible true values form $I(y) = [y - \varepsilon, y + \varepsilon]$. Acceptance follows when $I(y) \subseteq C$. Notice the change from a statement about one number to a statement about all numbers compatible with the information. Checking only $y \in C$ answers whether the displayed reading is acceptable. It does

not answer whether the unknown temperature is acceptable. If containment fails, the information allows an unacceptable temperature; it may also allow acceptable temperatures. The correct conclusion is that this acceptance test has not certified the chamber.

This distinction is useful whenever a calculation is an upper or lower bound rather than an exact observation. Write down what the bound encloses before deciding what its sign proves. A positive upper bound on an error does not mean that this error occurred. A negative lower bound on a safety margin does not mean the actual margin is negative. Both statements describe the limits of the information.

Choose before the disturbance arrives

Now consider one minute between commands. The constructed update is $x^+ = 1.1x + u + w$, where x^+ is the next temperature deviation, $u \in [-1, 0]$ is a commanded cooling contribution in degrees Celsius per step, and $w \in [0, 0.4]$ is an unknown warming contribution in the same units. The coefficient 1.1 is dimensionless. This simplified model assumes that the current x is known exactly, the command is realized exactly, and the stated disturbance interval contains the actual disturbance. Sensor uncertainty would require another universal quantifier over the possible current temperatures.

The command is selected after observing x and before observing w. Therefore the desired one-step statement is:

$$\forall x \in C \; \exists u \in [-1, 0] \; \forall w \in [0, 0.4] : \quad 1.1x + u + w \in C.$$

Read this aloud as a sequence of information: show me an allowed temperature; I choose a command; any allowed disturbance may then occur. If u were allowed to depend on w, the calculation would describe a different device with information available before the action. Mathematical punctuation is recording a physical timing assumption.

Worked example — A command interval at one measured temperature ▾

Decision. At $x = 4.5$ degrees Celsius, find every command that keeps the next step in C, whatever allowed warming occurs. Since the update increases with w, its complete range is $[4.95 + u, 5.35 + u]$. There is no need to test infinitely many disturbances separately: the two endpoints bound every intermediate value.

Translate containment. The lower endpoint must be at least zero and the upper endpoint at most five. Thus $u \geq -4.95$ and $u \leq -0.35$. Intersect these conditions with the actuator interval $[-1, 0]$ to obtain $u \in [-1, -0.35]$. The intersection matters: a mathematically useful command is irrelevant if the actuator cannot issue it.

Select and check. Choose $u = -0.5$. The successor interval is $[4.45, 4.85]$, contained in $[0, 5]$. The nominal disturbance $w = 0$ would predict 4.45; accounting for every allowed disturbance raises the upper prediction to 4.85. This verifies one step from the stated temperature. It has not yet verified a whole trajectory or a chamber with unmodeled delays.

To cover every $x \in C$, one explicit choice is $u(x) = -0.18x$. This command lies in $[-0.9, 0] \subseteq [-1, 0]$, and the resulting successor interval is $[0.92x, 0.92x + 0.4]$. Its lower endpoint is nonnegative; its upper endpoint is at most $0.92(5) + 0.4 = 5$. We have supplied a witness for the existential quantifier at every state, using one formula rather than infinitely many separate calculations.

No single fixed command works for the entire interval. At $x = 0$, allowing $w = 0$ forces $u \geq 0$, hence $u = 0$. At $x = 5$, allowing $w = 0.4$ forces $u \leq -0.9$. These requirements conflict. Feedback changes the available decision because the command may depend on the observed temperature. This conclusion follows from the two boundary states, not from how many points were simulated.

Extend the claim one step at a time

Suppose the same update and disturbance bounds apply at every integer time $t \geq 0$, and apply the feedback $u_t = -0.18x_t$. To prove $x_t \in C$ for all time, first state the base case $x_0 \in C$. Next assume only that the current x_t lies in C. The one-step calculation then shows $x_{t+1} \in C$ for every permitted w_t. Induction completes the argument. It permits disturbances to change arbitrarily from step to step, because the one-step result did not require them to be constant or independent.

The proof is conditional. If a disturbance exceeds 0.4, if the command arrives late, or if the temperature is not known as assumed, a hypothesis has changed. The proof remains mathematically valid while its applicability to that modified situation needs new work. Keeping hypotheses beside the step where they are used makes that limitation understandable rather than an unexplained disclaimer at the end.

A limit does not automatically give a stopping time

A controller may run a numerical calculation before issuing its command. Let e_k be a nonnegative temperature prediction error bound after iteration k. An error recurrence describes how an earlier

error is reduced and how new approximation error is introduced. The geometric factor explains the improvement; the additive term explains why unlimited iteration can still leave a nonzero bound.

Worked example — Transfer from a recurrence to a computation deadline

Model and goal. Suppose $e_0 = 0.8$ degrees Celsius and $e_{k+1} \le 0.6e_k + 0.02$. Each iteration costs 2 milliseconds and initialization costs 5 milliseconds. We require a certified error at most 0.06 degrees Celsius before issuing a command. The recurrence and timings are stipulated teaching assumptions.

Unroll the recurrence. The first bounds are $e_1 \le 0.5$ and $e_2 \le 0.32$. Repeating substitution gives $e_k \le 0.8(0.6)^k + 0.02\sum_{j=0}^{k-1}(0.6)^j$. Multiplying the finite sum by $1 - 0.6$ cancels adjacent powers, so the sum is $(1 - 0.6^k)/0.4$. Therefore:

$$e_k \le 0.05 + 0.75(0.6)^k.$$

Solve the actual tolerance. We need $0.75(0.6)^k \le 0.01$, equivalently $(0.6)^k \le 1/75$. Taking logarithms gives $k \ge \log(1/75)/\log(0.6)$; division reverses the inequality because $\log(0.6)$ is negative. Check the neighboring integers: at $k = 8$ the bound is 0.06259712, and at $k = 9$ it is 0.057558272. Nine iterations suffice, costing $5 + 9(2) = 23$ milliseconds. Eight are not certified by this bound.

Interpret the limit. The bound approaches 0.05, so a demand of 0.04 cannot be established from this recurrence. A demand of exactly 0.05 is also not reached by this finite bound. Actual errors might be smaller, but this information alone cannot certify them. Improving the additive approximation term could be more useful than buying more iterations.

COMMON MISTAKE — PROVING THE NOMINAL PROBLEM

Using $w = 0$ in the chamber or dropping 0.02 from the error recurrence makes the arithmetic easier while changing the claim. Repair the argument by writing the range of every unknown contribution and retaining it through each inequality. A successful nominal trajectory and a convergent geometric term are both useful calculations; each becomes a robust conclusion only after every remaining term is accounted for.

Application practice: information, action and guarantees

▼ Exercise 0.B1 — Easy: Accept a reading with a bounded error

A chamber must have $x \in [0, 5]$ degrees Celsius above reference. A sensor satisfies $|x - y| \leq 0.2$. Assess readings $y = 4.7$ and $y = 4.9$. Derive the complete interval of readings this rule can accept, and explain whether rejecting a reading proves physical failure.

Review the relevant tools

▼ Show hint

Write the possible true-temperature interval and require both its endpoints to satisfy the specification.

▼ Show worked solution

At 4.7, possible values are $[4.5, 4.9]$, all acceptable. At 4.9, the interval is $[4.7, 5.1]$, so acceptance is not certified. The second interval includes both passing and failing temperatures.

For a general reading, containment requires $y - 0.2 \geq 0$ and $y + 0.2 \leq 5$, giving $y \in [0.2, 4.8]$. This interval is closed because the specification allows its endpoints. A rejected reading indicates insufficient evidence under this rule; it does not identify the actual temperature. The conclusion also relies on the sensor error bound holding for the reading in question.

▼ Exercise 0.B2 — Medium: Keep the decision ahead of the unknown warming

Use $x^+ = 1.1x + u + w$, $u \in [-1, 0]$, $w \in [0, 0.4]$, and $C = [0, 5]$. At $x = 4.8$, find every robustly allowed command and verify $u = -0.8$. A colleague chooses $u = -0.28 - w$. Explain why that calculation uses different information, even if its numerical values lie in the actuator interval.

Review the relevant tools

▼ Show hint

First fix one u and range over w. Then ask when the colleague learns w.

▼ Show worked solution

The successor interval for fixed u is $[5.28 + u, 5.68 + u]$. Containment requires $u \geq -5.28$ and $u \leq -0.68$; intersecting with the actuator interval gives $[-1, -0.68]$. At $u = -0.8$ the successor

interval is $[4.48, 4.88]$, so every allowed disturbance passes.

The colleague's command gives $x^+ = 5$ for each disturbance, and ranges from -0.68 to -0.28. But it requires knowing w before commanding: it witnesses $\forall w\ \exists u$, whereas the device requires $\exists u\ \forall w$. A command issued after measuring the disturbance would be a different timing model. The robust interval above supplies the required single command.

▼ Exercise 0.B3 — Medium: Recognize an attainable tolerance

An iterative predictor satisfies $e_0 = 0.70$ and $e_{k+1} \le 0.5e_k + 0.03$, in degrees Celsius. Derive a bound for every k. Can it certify $e_k \le 0.04$? Find the smallest iteration budget certified for $e_k \le 0.07$ and explain how the answer changes for the strict requirement $e_k < 0.07$.

Review the relevant tools

▼ Show hint

The steady value of the equality recurrence solves b=0.5b+0.03. Subtract it before taking powers.

▼ Show worked solution

The steady value is $b = 0.06$. Subtracting it gives $e_{k+1} - 0.06 \le 0.5(e_k - 0.06)$; induction yields $e_k \le 0.06 + 0.64(0.5)^k$. The requested 0.04 is below the limiting bound and cannot be certified by this information.

For 0.07, require $0.64/2^k \le 0.01$, or $2^k \ge 64$. The smallest integer is $k = 6$, where the bound equals 0.07; at $k = 5$ it is 0.08. Strict inequality needs $k = 7$, giving 0.065. Because equality at every step is an allowed error sequence, the failed smaller budgets really cannot provide the corresponding universal guarantee.

▼ Exercise 0.B4 — Hard: Find exactly how much disturbance the actuator can cover

Replace the warming bound by $w \in [0, d]$, with $d \ge 0$, in the chamber model. For which d does $\forall x \in [0, 5]\ \exists u \in [-1, 0]\ \forall w \in [0, d] : x^+ \in [0, 5]$ hold? Prove necessity with a boundary state and sufficiency with an explicit feedback. Then state the all-time consequence and its initial-state assumption.

Review the relevant tools

▼ Show hint

At x=5 test the largest warming and strongest available cooling. For sufficiency try u=-0.2x.

▼ Show worked solution

At $x = 5, w = d$, even maximum cooling gives $x^+ = 5.5 - 1 + d = 4.5 + d$. If $d > 0.5$, this exceeds five, so no command can satisfy the claim. Thus $d \leq 0.5$ is necessary.

For any $0 \leq d \leq 0.5$, choose $u = -0.2x$, which lies in $[-1, 0]$. Then $x^+ = 0.9x + w$ is nonnegative and at most $4.5 + d \leq 5$. This proves sufficiency for the entire interval, including its boundary, and establishes the exact parameter range $[0, 0.5]$.

If $x_0 \in [0, 5]$, applying this feedback repeatedly keeps every subsequent state in that interval by induction, for arbitrary permitted disturbance sequences. Starting outside the interval is not covered. Notice that the feedback used earlier is a valid choice for $d = 0.4$, while this new feedback also handles the limiting case $d = 0.5$.

Reconstruct the argument before moving on

Close the examples and reconstruct the chain in your own words: specification, information set, command timing, one-step containment, induction, and computation budget. At each link, name a hypothesis that could break its application. Explain why a failed sufficient test differs from a counterexample, and why a limiting value differs from a finite stopping guarantee. These distinctions let you read later safety statements without silently strengthening what they prove.

For recall, write the negation of the robust chamber claim; identify which variable may depend on which earlier variables; and explain where the initial condition enters the induction. Then derive the error floor for a general recurrence $e_{k+1} \leq ae_k + b$ with $0 \leq a < 1$ and $b \geq 0$. The answer $b/(1 - a)$ should follow from the same finite geometric sum, rather than from memory alone.

The next bridge is geometric. Primer A's vectors and norms describe information sets when several temperatures or positions are uncertain together. Primer B's constrained optimization selects a good command from the admissible set. The logic here continues to determine what either calculation is allowed to conclude.

Exercises

For gradual practice on the opening material, use the graded sets on sets & logic, functions & fixed points, and proofs. The exercises below bring together ideas from across the primer.

▼ Exercise 0.1 — Negation and quantifier order: feedback versus one fixed input

Let $C = U = [-1, 1]$ and $f(x, u) = 2x + u$. (a) Negate "$\forall x \in C\ \exists u \in U : f(x, u) \in C$". (b) Prove that statement. (c) Prove that "$\exists u \in U\ \forall x \in C : f(x, u) \in C$" is false. (d) Interpret the difference.

▼ Show answer

(a) $\exists x \in C\ \forall u \in U : f(x, u) \notin C$: some state of C is lost whatever input is applied. (b) Given $x \in C$, take $u = -x \in U$; then $f(x, u) = x \in C$. (c) Suppose one u works for all $x \in C$. $x = 1$ needs $2 + u \leq 1$, so $u \leq -1$; $x = -1$ needs $-2 + u \geq -1$, so $u \geq 1$: a contradiction. (d) In "$\forall x\ \exists u$" the input may depend on the state: the feedback $u = -x$ makes C invariant (control invariance). "$\exists u\ \forall x$" asks for one open-loop input that works everywhere, and none exists.

▼ Exercise 0.2 — A monotone set operator: reachability is fragile

Redo the worked example of Section 2 ($D = \{0, \ldots, 8\}$, $L = 1$, $h = 0$, $f = (1.8,\ 2.3,\ 1.4,\ 2.4,\ 1.4,\ 2.4,\ 1.4,\ 0.4,\ 1.4)$, $S_0 = \{0\}$) with estimation error $\epsilon = 0.5$: $R_\epsilon(S) = S \cup \{x : \exists x' \in S : f(x') - \epsilon - |x - x'| \geq 0\}$. (a) Compute the iterates and $\bar{R}_\epsilon(S_0)$. (b) Compare the number of strict increases with the Fact's bound. (c) Why does the closure shrink from 8 points to 3?

▼ Show answer

(a) From $x' = 0$: $1.3 \geq |x|$ certifies $\{0, 1\}$. From $x' = 1$: $1.8 \geq |x - 1|$ certifies $\{0, 1, 2\}$ (point 3 would need $2 \leq 1.8$). From $x' = 2$: $0.9 \geq |x - 2|$ certifies only $\{2\}$. So $S_1 = \{0, 1\}$, $S_2 = S_3 = \{0, 1, 2\} = \bar{R}_{0.5}(S_0)$. (b) At most $|D| - |S_0| = 8$ strict increases are allowed; two occur. (c) With $\epsilon = 0$ the gateway point 3 was certified from $x' = 1$ with slack $2.3 - 2 = 0.3 < 0.5$, and the dip at 2 cannot help ($1.4 - 0.5 - 1 = -0.1 < 0$): everything behind point 3 is lost. The reachable set jumps as ϵ varies.

▼ Exercise 0.3 — Induction and discounted sums

(a) Prove by induction that $\sum_{t=0}^{T-1} \gamma^t = \frac{1-\gamma^T}{1-\gamma}$ for $\gamma \neq 1$. (b) For $0 \leq \gamma < 1$, show $\sum_{t \geq 0} t\gamma^t = \frac{\gamma}{(1-\gamma)^2}$ by writing $t = \sum_{s=1}^{t} 1$ and exchanging the order of summation; why is that allowed? (c) Evaluate for $\gamma = 0.9$ and interpret $(1-\gamma)\sum_t t\gamma^t$.

▼ Show answer

(a) $T = 0$: the empty sum is $0 = \frac{1-1}{1-\gamma}$. Step: $\sum_{t=0}^{T} \gamma^t = \frac{1-\gamma^T}{1-\gamma} + \gamma^T = \frac{1-\gamma^T+\gamma^T-\gamma^{T+1}}{1-\gamma} = \frac{1-\gamma^{T+1}}{1-\gamma}$. (b) $\sum_{t \geq 0}\sum_{s=1}^{t} \gamma^t = \sum_{s \geq 1}\sum_{t \geq s} \gamma^t = \sum_{s \geq 1} \frac{\gamma^s}{1-\gamma} = \frac{\gamma}{(1-\gamma)^2}$, using the tail formula twice; the exchange is allowed because all terms are nonnegative. (c) $0.9/0.01 = 90$. Since $(1-\gamma)\gamma^t$ are probabilities summing to 1, $(1-\gamma)\sum_t t\gamma^t = \frac{\gamma}{1-\gamma} = 9$ is the mean of a random time with $\mathbb{P}(\tau = t) = (1-\gamma)\gamma^t$: discounting acts like a random horizon of mean 9 steps.

▼ Exercise 0.4 — Suprema, limsup and the max-min inequality

(a) Find $\inf$, $\min$, $\sup$, $\max$ of $A = \{x/(1+x) : x \geq 0\}$ where they exist. (b) For $a_t = (-1)^t\big(1 + \frac{1}{t+1}\big)$ find $\sup$, $\inf$, $\limsup$, $\liminf$. (c) Compare $\sup_x(\sin x + \cos x)$ with $\sup \sin + \sup \cos$. (d) For $\phi(x, y) = xy$ on $[-1, 1]^2$, compute $\inf_x \sup_y \phi$ and $\sup_y \inf_x \phi$.

▼ Show answer

(a) $x/(1+x) = 1 - \frac{1}{1+x}$ increases from 0 towards 1: $\inf = \min = 0$, $\sup = 1$, no maximum. (b) $a_0 = 2, a_1 = -1.5, a_2 = \frac{4}{3}, a_3 = -1.25, \ldots$; even terms decrease to 1, odd terms increase to -1. So $\sup = \max = 2$, $\inf = \min = -1.5$, $\limsup = 1$, $\liminf = -1$. (c) $\sin x + \cos x = \sqrt{2}\sin(x + \pi/4)$, so the supremum is $\sqrt{2} \approx 1.414 < 2$: the two suprema are attained at different points. (d) $\sup_y xy = |x|$, so $\inf_x \sup_y \phi = 0$; $\inf_x xy = -|y|$, so $\sup_y \inf_x \phi = 0$. No gap here: $(0, 0)$ is a saddle point.

▼ Exercise 0.5 — Does a minimizer exist?

Decide with a reason, and compute where possible: (a) $\min_{x \geq 0} e^{-x}$; (b) $\min_{x \in [0,1]} (x^2 - x)$; (c) $\min \|u\|^2$ subject to $u_1 + u_2 \geq 1$ in $\mathbb{R}^2$; (d) $\max_{u \in \mathbb{R}^2} (u_1 + u_2)$; (e) for which c is $\{x \in \mathbb{R} : x^2/(1+x^2) \leq c\}$ compact?

▼ Show answer

(a) No: $e^{-x} > 0 = \inf$ on an unbounded domain. (b) Yes (compact, continuous): $2x - 1 = 0$ gives the minimum $-\frac{1}{4}$ at $x = \frac{1}{2}$. (c) Yes (closed nonempty feasible set, coercive objective): $\|u\|^2 \geq \frac{1}{2}(u_1 + u_2)^2 \geq \frac{1}{2}$ because $(u_1 - u_2)^2 \geq 0$, with equality at $u^* = (\frac{1}{2}, \frac{1}{2})$. (d) No: $u = (s, s)$ gives $2s \to \infty$. (e) The function is below 1 everywhere; for $0 \leq c < 1$ the set is $\{x^2 \leq \frac{c}{1-c}\}$, a closed bounded interval, while for $c \geq 1$ it is all of $\mathbb{R}$ (and for $c < 0$ it is empty). So the set is compact exactly for $c < 1$ (the empty set counts as compact); $c \geq 1$ fails because the function tends to 1 as $|x| \to \infty$.

▼ Exercise 0.6 — Reading rates and costs

(a) Order $(\ln T)^3$, $\sqrt{T}$, $T^{2/3}$, $T/\ln T$ by growth and justify two comparisons. (b) Is $0 \leq R_T \leq 4\sqrt{T}\ln T$ sublinear, and from which T on is the average-regret bound at most 0.1? (c) A Cholesky factorisation takes 0.05 s at size 1000; estimate size 4000. (d) How many points has a grid with 20 points per axis in $d = 6$?

▼ Show answer

(a) $(\ln T)^3 \ll \sqrt{T} \ll T^{2/3} \ll T/\ln T$, e.g. $\sqrt{T}/T^{2/3} = T^{-1/6} \to 0$ and $T^{2/3}/(T/\ln T) = \ln T / T^{1/3} \to 0$. (b) $0 \leq R_T/T \leq 4\ln T/\sqrt{T} \to 0$: sublinear. This bound decreases for $T \geq e^2$, is approximately 0.368 at $T = 10^4$ and 0.146 at 10^5, and is at most 0.1 from $T = 246{,}639$ on. (c) $O(t^3)$: $4^3 = 64$ times the work, about 3.2 s (a rough estimate). (d) $20^6 = 6.4 \cdot 10^7$; for $d = 10$ it would be $20^{10} = 1.024 \cdot 10^{13} \approx 1.0 \cdot 10^{13}$.

▼ **Exercise 0.7 — Banach's theorem for $\cos x$: the bound versus reality**

(a) Show that $g(x) = \cos x$ maps $J = [\cos 1, 1]$ into itself and is a contraction there with $L = \sin 1$. (b) From $x_0 = 1$, how many iterations does the a priori bound guarantee for $|x_k - x^*| \le 10^{-6}$? (c) In fact 32 suffice; explain the gap. (d) Evaluate the a posteriori bound at $k = 10$, where $x_9 \approx 0.731404$, $x_{10} \approx 0.744237$.

▼ **Show answer**

(a) $\cos$ decreases on $[0, \pi/2] \supseteq J$, so $\cos(J) = [\cos 1, \cos(\cos 1)] \subseteq J$, with endpoints approximately 0.5403 and 0.8576. By the mean value theorem $|\cos x - \cos y| = |\sin \xi|\,|x - y| \le \sin 1\,|x - y|$ on J, so $L = \sin 1 \approx 0.8415$; J is closed, and Banach gives the unique fixed point $x^* \approx 0.739085$. (b) $|x_1 - x_0| = 1 - \cos 1 \approx 0.4597$ and $1 - L = 1 - \sin 1 \approx 0.1585$. The exact a priori certificate is equivalent to $k \ge \ln\big((1 - \cos 1)/((1 - \sin 1)10^{-6})\big)/\ln(1/\sin 1) \approx 86.2$, so its first integer cutoff is 87. For the rounded inputs, $\ln\big(0.4597/(0.1585 \cdot 10^{-6})\big) \approx 14.88$ and $\ln(1/0.8415) \approx 0.1726$. (c) L is the worst slope on J, but near x^* the slope is $\sin x^* \approx 0.674$; with $|x_0 - x^*| \approx 0.261$ the local-rate estimate $\ln(0.261 \cdot 10^6)/\ln(1/0.674) \approx 31.6$ suggests 32. A direct certified iteration verifies that 32 suffice. The global bound must hold for every start in J. (d) $\frac{L}{1-L}|x_{10} - x_9| \approx 5.31 \cdot 0.012833 \approx 0.068$, against the true error ≈ 0.0052 and the a priori bound ≈ 0.52 at $k = 10$.

Ready to move on?

You are ready for Primer A when you can read and negate a quantified statement, compose two functions, and explain the difference between a bound and an attained optimum. Check these without opening a solution:

- Redo negating quantifiers, composition, and induction.
- Redo a discounted tail and supremum versus maximum.
- Explain why the grid margin is needed, then turn the explicit error bound into a budget.

If one task stalls, revisit its review link and attempt the local Easy exercise again. You can continue once Easy and Medium work is independent; the Hard exercises are useful return visits as later modules give the ideas context. Next: Primer A: vectors and matrix tools.

Further Reading

Resource	Edition	Covers	Why read it
Book of Proof (R. Hammack)	3rd ed. 2018, free	Sets, logic, proof techniques, induction, functions	The gentlest complete route through Sections 1–3.
How to Prove It: A Structured Approach (D. J. Velleman)	3rd ed., CUP 2019	Quantifiers, proof strategies, induction	How the logical form of a statement dictates its proof.
Mathematics for Computer Science (E. Lehman, F. T. Leighton, A. R. Meyer)	MIT 6.042J, 2015, free	Logic, induction and state-machine invariants, sums, asymptotic notation	Invariants are induction over time steps; its "Sums and Asymptotics" matches Sections 4 and 7.
Understanding Analysis (S. Abbott)	2nd ed., Springer 2015	Suprema, sequences, series, compactness, continuity	The most readable treatment of Sections 4–6.
Basic Analysis I (J. Lebl)	v6.3, 2026, free	limsup, series, extreme values, uniform continuity, metric spaces	Free and complete; Section 7.6 proves the contraction fixed-point theorem.
Convex Optimization, App. A (S. Boyd, L. Vandenberghe)	CUP 2004, free PDF	Norms, open and closed sets, sup and inf	A compact reference in the optimization modules' conventions.
Computational Complexity: A Modern Approach (S. Arora, B. Barak)	CUP 2009, free draft	P, NP, reductions (Chapter 2)	The precise meaning of "NP-hard" in Modules 12 and 15.
Reinforcement Learning: An Introduction (R. S. Sutton, A. G. Barto)	2nd ed., MIT Press 2018, free	Discounted returns, value iteration	Where the geometric series and fixed points meet RL.

Flashcards

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Negate "$\forall x \in C\ \exists u \in U : f(x,u) \in C$". How does it differ from "$\exists u\ \forall x$"?

Show answer

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